

OPEN INSTITUTIONAL COMPILER

Technical Design Document

Owner-approved design baseline, pre-external-review, for the open reference implementation, hosted Institutional Compiler Lab, OIC-Bench, public claims, and capability roadmap.

Status: OWNER-APPROVED DESIGN BASELINE - PRE-EXTERNAL-REVIEW

Effective date: 28 July 2026

Design authority: Arkadiy Miteiko / Veraxis

Revision basis: Supersedes TDD-OIC-001 v1.0; incorporates External Review ER-OIC-001

Immediate implementation horizon: Seven-day experimental reference vertical slice

Evidence horizons: Thirty-day reproducible alpha; sixty-day full benchmark and external review; ninety-day enterprise pilot candidate

Document Control

This document is the owner-approved technical and product design baseline for the Open Institutional Compiler (OIC), before completion of independent technical, security, legal, benchmark, ZTL, and VEIP review. It authorizes bounded exploratory implementation against the stated interfaces and invariants. It does not represent standards-body approval, customer adoption, independent validation, legal certification, enterprise-production readiness, or evidence of market-wide novelty.

The document is intended to make the design reconstructable and falsifiable by engineers, policy practitioners, security reviewers, benchmark evaluators, open-source contributors, and enterprise buyers who were not present during its creation. It is deliberately explicit about what is demonstrated, proposed, borrowed, project-controlled, unverified, and forbidden as a public claim.

Field	Value
Document ID	TDD-OIC-001
Version	1.1
Status	OWNER-APPROVED DESIGN BASELINE - PRE-EXTERNAL-REVIEW
Effective date	28 July 2026
Owner / design authority	Arkadiy Miteiko / Veraxis
Supersedes	TDD-OIC-001 v1.0
Revision evidence	External Review ER-OIC-001; revision disposition recorded in Appendix H
Governed artifacts	Owner-authorized exploratory implementation of the open GitHub repository, hosted Institutional Compiler Lab, OIC-Bench, public claims register, and capability roadmap
Current authority	Engineering may implement the seven-day experimental vertical slice. Release-freeze, enterprise-readiness, comparative-superiority, and standards-like claims remain blocked by the gates in Sections 10, 18, and 19.
Change control	Material architectural, semantic, benchmark, security, claims, or dependency changes require an Architecture Decision Record and updated traceability.
Approval limitation	Owner approval authorizes implementation. It does not constitute legal, regulatory, standards-body, customer, or independent technical certification.

Status vocabulary

- **PROPOSED:** Design exists but is not authorized for implementation.
- **UNDER REVIEW:** Design is being evaluated; interfaces and thresholds may change.
- **OWNER-APPROVED DESIGN BASELINE - PRE-EXTERNAL-REVIEW:** Owner-authorized implementation may proceed, but required independent reviews and release evidence are incomplete.
- **APPROVED FOR IMPLEMENTATION:** Required design reviewers have accepted the implementation contracts for the declared scope.
- **FROZEN FOR RELEASE:** Implementation, dependency dossiers, benchmark, claims, security evidence, and release artifacts are pinned.
- **SUPERSEDED:** Retained for history but no longer controlling.

Normative language

The terms **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** are normative only within the owner-authorized OIC development process. Their use does not imply adoption by an external standards body. "Institution" means the organization whose documents, authority, review decisions, and runtime consequences are being compiled. "Reviewer" means a person operating under a recorded institutional role; it does not imply legal competence unless separately established.

Governing design rule

Every requirement **MUST** trace to a customer problem. Every architectural choice **MUST** trace to an evaluated alternative. Every quality claim **MUST** trace to a pinned benchmark. Every runtime result **MUST** trace to an admitted control, interpretation, source, and current authority state. Every schedule, dependency, cost, and novelty claim **MUST** disclose whether it is measured, sourced, estimated, proposed, or unverified.

Founder design thesis - non-normative

The following statement expresses the long-term motivation for the architecture. It is not a technical requirement, approval claim, benchmark result, or release gate:

"May every generation leave civilization with a greater capacity to observe, justify, and correct the institutional consequences it creates than the generation that came before."

Revision 1.1 Disposition

Version 1.1 preserves the core architecture of v1.0 while correcting five evidence and governance defects identified by External Review ER-OIC-001: an implausibly overloaded seven-day schedule; unverified project-controlled ZTL and VEIP dependencies; a maturity label stronger than the completed approvals; absence of a commercial competitive landscape; and numerical specificity without evidence classification. It also separates anticipated objections from actual external dissent and moves aspirational framing outside the normative technical register.

Review finding	Disposition in v1.1	Remaining evidence gap
Seven-day plan overloaded with full benchmark, usability, security, and enterprise claims	Split into 7-, 30-, 60-, 90-day and post-pilot evidence horizons; week one uses a named benchmark preflight subset	Actual velocity and integration performance remain unmeasured
ZTL and VEIP treated like externally mature dependencies	Reclassified as project-controlled components with provisional adapters and mandatory verification dossiers	Public/pinned artifacts, licenses, interfaces, conformance evidence, and independent reproduction must be completed
"Governing" status borrowed standards-body maturity	Replaced with owner-approved, pre-external-review status	External technical, security, legal, benchmark, ZTL, and VEIP approvals remain pending
No commercial landscape	Added public-materials capability comparison covering GRC, regulatory intelligence, legal AI, and runtime AI governance vendors	No hands-on product evaluation or confidential vendor diligence has been performed
Targets and cost figures appeared measured	Added evidence-classification requirements and relabeled figures as proposed gates or planning estimates	Calibration requires benchmark annotation, baseline runs, and measured delivery data
Dissent was simulated	Renamed internally generated material to anticipated objections; ER-OIC-001 entered as first external review record	Additional named external reviewers are required before release freeze
Technical and aspirational registers were mixed	Founder thesis moved to a clearly non-normative note	Public narrative still requires claims review before launch

The architectural baseline remains a design hypothesis. Its claimed commercial and technical differentiation becomes evidence only when the open repository, hosted Lab, frozen benchmark, external reviews, and independent use make the relevant claims reproducible.

Executive Technical Decision

The specific customer problem is upstream of runtime admission:

How does a regulated enterprise get from its existing documents to something its runtime can actually enforce?

Regulated enterprises hold the material that governs consequential work in policies, manuals, approval matrices, procedures, amendments, definitions, prior findings, local interpretations, templates, contracts, and institutional practice. Runtime systems require explicit actors, actions, authority limits, conditions, exceptions, evidence obligations, time boundaries, escalation paths, and decision behavior. The conversion between these forms is presently labor-intensive, repeatedly reinterpreted, weakly versioned, and often detached from source authority.

OIC is designed to cross that gap. It transforms a bounded institutional corpus into source-anchored candidate meaning, exposes ambiguity and conflict, records scoped institutional review, produces a portable Open Control Envelope, compiles that envelope to runtime targets, and preserves lineage from source through execution, reliance, and correction.

OIC is not ordinary policy-as-code. Policy-as-code usually begins after rules are sufficiently explicit to encode. OIC addresses the preceding institutional compilation process by which fragmented paperwork becomes a reviewable and enforceable object.

The design is expressed through two artifacts:

1. **Open reference implementation on GitHub.** It is intended to prove inspectability, local reproducibility, extensibility, and independence from a hosted semantic service.
2. **Institutional Compiler Lab.** It is intended to prove that legal, compliance, audit, risk, security, and enterprise-architecture practitioners can inspect a bounded corpus, resolve a prioritized docket, admit scoped interpretations, compile envelopes, and test runtime behavior through a low-training interface.

Both artifacts **MUST** use the same open semantic compiler core. The hosted platform **MAY** add identity, tenancy, secure storage, orchestration, telemetry, and managed operations. It **MUST NOT** hide a materially different semantic transformation.

Current evidence status

Proposition	Current status at v1.1
Customer problem exists	Strong design hypothesis supported by established policy-to-control workflows; customer baseline study not yet completed
Twelve-module architecture is implementable	Owner-approved architecture; vertical-slice integration unproven
Open Control Envelope is technically differentiating	Plausible design hypothesis; commercial landscape review added, but market-wide novelty is not proven
OIC outperforms direct document-to-code systems	Unproven until frozen benchmark and baselines run
Fortune 500 practitioners can use the Lab with a light education curve	Unproven until a real usability study is completed
ZTL and VEIP can serve as reliable load-bearing components	Provisional; verification dossiers and pinned interfaces are release blockers
Seven days is sufficient	Only for the experimental reference slice defined in Section 16; not for full OIC-Bench, enterprise readiness, or superiority claims

Product promise

Upload a bounded institutional policy corpus. Receive source-anchored candidate controls, a prioritized review docket, scoped admission records, a portable control envelope, executable policy artifacts, conformance tests, and source-to-runtime lineage within days rather than months.

This is a target product promise. The first release may claim it only for the named reference corpus and measured scope.

Novel technical object

The central abstraction is the **institutionally admitted Open Control Envelope**: a portable, target-independent enforcement contract whose fields remain bound to source anchors, authority basis, interpretation status, reviewer disposition, evidence requirements, semantic version, test suite, effective period, and correction path.

The system separates four acts that many existing workflows collapse:

1. **Extraction:** What propositions and structures appear in the source?
2. **Interpretation:** What operational meanings are plausible?
3. **Admission:** Which interpretation has an authorized institution accepted, for what scope and use?
4. **Enforcement:** How is the admitted meaning projected into a target runtime without reinterpretation?

The object is novel as an OIC design hypothesis. The repository and benchmark must establish whether the abstraction is independently useful, consumed, and differentiated from existing proprietary control representations.

Delivery and evidence horizons

Horizon	Purpose	Evidence state permitted
Seven-day experimental reference slice	Prove one bounded end-to-end path from a prepared difficult corpus to a live runtime result	Experimental tag; architecture and integration evidence only; no enterprise-readiness or superiority claim
Thirty-day reproducible alpha	Harden interfaces, dependency dossiers, open reproduction, security preflight, benchmark tooling, and hosted Lab alpha	Reproducibility and bounded capability claims only
Sixty-day evidence release	Complete OIC-Bench v0.1, named baselines, practitioner study, external technical/security review, and public limitations	Corpus-scoped benchmark claims if all frozen gates pass
Ninety-day enterprise pilot candidate	Prove controlled private deployment, tenancy, SSO, on-prem profile, operational controls, and change propagation with a design partner	Pilot-readiness claims for the tested profile; not universal production readiness
Post-pilot maturity	Independent corpus use, external PRs/adapters, repeat deployments, correction evidence, and independent benchmark reproduction	Category and infrastructural-maturity claims may be considered through the claims ledger

Decision summary

Decision	Selected approach	Evidence/status
Core architecture	Admitted canonical Institutional IR plus portable Open Control Envelope	Owner-approved design hypothesis
Implementation form	Modular monolith for the reference build; explicit immutable module contracts	Selected for delivery speed and failure isolation
Primary execution target	OPA/Rego	Public mature dependency; pin and conformance-test required
Secondary representation	DMN export for suitable decision logic	Deferred until Rego reference path is stable
Logical warrant	Provisional ZTL adapter	Project-controlled dependency; verification dossier required before release freeze
Execution continuity	Provisional VEIP adapter	Project-controlled dependency; verification dossier required before release freeze
Document ingestion	Docling adapter behind OIC contract	Reused public component; sandbox and version pin required
Hosted platform	Same open compiler core plus managed delivery shell	No-hidden-compiler invariant
Default failure behavior	No silent ALLOW	Missing support, authority, admission, evidence, current version, or facts must block, escalate, or return CANNOT

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1. Customer Problem, Why Now, and Cost of Inaction

1.1 Specific customer problem

Regulated enterprises cannot reliably translate the human-readable material that governs consequential work into explicit controls their software and agents can enforce. The bottleneck is not the absence of runtime gates. The bottleneck is producing a coherent, reviewable control envelope from heterogeneous institutional sources without silently losing authority, exceptions, temporal scope, evidence obligations, uncertainty, or correction paths.

A policy document does not arrive as a clean predicate such as:

- actor A may perform action B;
- under mandate C;
- within scope D;
- while conditions E through G remain true;
- subject to exception H;
- with evidence I;
- and escalation path J.

It arrives as prose distributed across statutes, contracts, policies, manuals, annexes, definitions, procedures, amendments, approval matrices, prior findings, review notes, templates, and informal practice. The sources may overlap, conflict, age differently, refer to undefined terms, or use the same term inconsistently. Existing runtime tools generally assume this upstream semantic and institutional work has already been completed.

1.2 Customer segments

The primary users are not compiler researchers. They are practitioners accountable for converting institutional requirements into operations:

- enterprise legal operations and legal engineering;
- compliance policy owners and control-design teams;
- internal audit and assurance functions;
- enterprise risk and operational resilience teams;
- security architecture, IAM, and policy-engine teams;
- procurement and payment-control owners;
- responsible AI and model-risk teams;
- enterprise architects integrating agent platforms;
- government digital services and regulated public institutions.

Secondary users include policy-engine maintainers, formal-methods engineers, rules-as-code practitioners, document-processing contributors, open-source security engineers, and domain experts contributing policy packs and benchmarks.

1.3 Current workflow and failure pattern

A typical enterprise currently performs a repeated manual chain:

1. A policy owner publishes or updates prose.
2. A business analyst creates a control matrix or requirements document.
3. A product or engineering team interprets the matrix for one application.
4. Developers encode logic in workflow configuration, IAM rules, service code, prompts, or a policy engine.
5. Testers derive examples from the implementation rather than the source.
6. Audit later reconstructs why the implementation behaved as it did.
7. A later amendment triggers another manual search for affected systems.

This workflow creates semantic duplication. Each system becomes a partially independent interpretation of the same institution. The enterprise may own the documents while losing control of the executable meaning that accumulates around them.

1.4 Why now

Several conditions have converged:

- Agentic systems can propose and execute consequential actions at a frequency that makes undocumented human interpretation unscalable.
- Mature policy engines can enforce structured controls, but they do not manufacture institutionally admitted controls from inherited documents.
- Modern document parsing and language models make source-anchored candidate extraction economically feasible, provided model output is treated as untrusted.

- Regulated enterprises increasingly require vendor portability, on-premises operation, and institution-owned governance artifacts.
- Model capability is becoming easier to rent; the scarce enterprise asset is the institution-specific meaning, authority, workflow, evidence, and correction logic that makes capability dependable.
- The cost of leaving policy interpretation inside prompts, application code, and vendor-specific configuration rises sharply as AI becomes an actor rather than an assistant.

1.5 Cost of not building

The cost of inaction MUST be measured during the reference and pilot phases. The baseline study will quantify:

- analyst hours to create a control matrix manually;
- engineering hours to implement and test each control;
- elapsed time from policy publication to enforceable deployment;
- time to trace a runtime rule to its governing source;
- unsupported executable-field rate;
- number of divergent interpretations across applications;
- amendment impact analysis time and missed dependencies;
- audit reconstruction time per consequential decision;
- incidents caused by stale, excessive, or undocumented authority;
- delayed agent deployments attributable to control uncertainty.

Until measured, cost figures are planning hypotheses, not public claims. The commercial hypothesis is that OIC reduces first-pass control production time, lowers unsupported implementation, improves amendment impact discovery, and creates reusable institution-owned enforcement artifacts.

1.6 Capability change that closes the gap

OIC closes the gap by introducing a controlled transformation:

Source corpus → anchored source graph → candidate normative meaning → authority and ambiguity review → institutional admission → canonical Institutional IR → logical warrant → portable control envelope → target policy → runtime decision → evidence, reliance, and correction.

The capability is complete only when a skeptical reviewer can move in both directions: from source to runtime behavior and from a runtime consequence back to the exact source, interpretation, reviewer disposition, control version, evidence, and warrant used.

2. Goals, Non-Goals, Claims, and Design Principles

2.1 Goals

- Compile bounded, heterogeneous enterprise policy corpora into source-anchored candidate normative units.
- Reconstruct document status, authority basis, temporal scope, precedence, delegation, and supersession sufficiently to support review.
- Expose ambiguity, conflict, missing definitions, missing delegation, and open-textured judgment rather than silently resolving them.
- Allow authorized nontechnical reviewers to admit, qualify, modify, reject, escalate, or mark interpretations non-automatable.
- Maintain a target-independent canonical Institutional IR and produce portable Open Control Envelopes.
- Compile admitted envelopes into OPA/Rego and JSON runtime contracts in the first release; support DMN export for suitable rules.
- Generate positive, negative, boundary, exception, unknown, expiry, revocation, and conflict tests.
- Integrate ZTL for logical warrant without conflating logic with source truth or institutional authority.
- Integrate VEIP for proposal, admission, execution, record, reliance, replay, revocation, and correction continuity.
- Deliver an open repository that runs locally and a hosted Lab using the same open semantic core.
- Publish a pinned benchmark that measures source fidelity, normative fidelity, authority fidelity, uncertainty integrity, behavior, change impact, usability, and operations.
- Disclose limitations and prevent unsupported public claims by design.

2.2 Non-goals

- Autonomously determine that an institution's interpretation is legally correct.
- Replace authorized legal, compliance, policy, audit, or regulatory review.
- Manufacture authority absent from the corpus or infer undocumented delegation from custom.
- Convert every open-textured standard into a Boolean predicate.
- Build a universal legal ontology or fully implement Akoma Ntoso, LegalRuleML, Catala, OpenFisca, or DMN.
- Build a new OCR engine, generic policy runtime, or workflow engine.
- Claim that extraction confidence creates institutional admission.
- Claim that a passing runtime test proves legal or regulatory compliance.
- Support every jurisdiction, language, document class, or policy engine in the reference release.
- Require the hosted platform for compilation, review, or evaluation.
- Allow confidential or regulated production data in the public Lab launch.
- Present OIC as finished infrastructure for every Fortune 500 workload after one week.

2.3 Claims ladder

Class	Meaning	Example
Release claim	Directly demonstrated by pinned release artifacts	OIC generates source-anchored candidate controls and compiles admitted deterministic controls to Rego.
Benchmark claim	Supported within a named frozen corpus and protocol	On OIC-Bench v0.1, unsupported executable-field rate was X% under protocol Y.

Class	Meaning	Example
Thesis claim	Future-looking architectural proposition	Dependable regulated agents require institution-owned controls compiled from governing sources.
Forbidden claim	Not supported and MUST NOT be published	OIC automatically guarantees legal compliance or eliminates expert review.

2.4 Design principles

Principle	Required consequence
Source before semantics	No candidate field exists without exact source support or an explicit derived/inferred label.
Admission before enforcement	A candidate interpretation cannot become enforceable merely because an LLM or compiler produced it.
Unknown is a first-class state	Missing or contested information never silently becomes false, true, or authorized.
Authority is not prose confidence	The probability that a model extracted a clause correctly does not establish the issuer's authority.
Target independence	Institutional meaning remains in the canonical IR; target code is a projection.
Human judgment is compilable without being automated	The compiler can encode who judges, what factors and evidence apply, and how review occurs even when it cannot compute the substantive answer.
No hidden compiler	Hosted and local artifacts share the same semantic transformation.
Fail closed, fail visible	Missing source, authority, admission, evidence, version, or facts yields DENY, CANNOT, ESCALATE, or NOT_APPLICABLE - never silent ALLOW.
Change is part of meaning	Effective dates, supersession, amendments, revocation, and correction are core semantics.
Open interfaces, replaceable components	Commodity parsers, model providers, graph stores, and target engines sit behind versioned OIC contracts.
Falsifiable claims	Every public quality claim maps to a frozen corpus, metric, threshold, and reproducible command.
Minimum education curve	A policy practitioner should review source and operational meaning without learning Rego, LegalRuleML, or formal logic.

2.5 System invariants

- I-01: No executable control field without at least one source anchor or an explicit institutionally admitted derivation record.
- I-02: No enforcement artifact from an interpretation_state earlier than admitted.
- I-03: Formula value T/F/Z is distinct from operational verdict ALLOW/DENY/CANNOT/ESCALATE/NOT_APPLICABLE.
- I-04: Unknown never serializes as grounded false or authorized allow.
- I-05: Expired, superseded, suspended, or revoked authority cannot yield ALLOW.
- I-06: Every admitted interpretation records reviewer identity, role, authority basis, scope, timestamp, and disposition hash.
- I-07: Every target artifact binds to IR version, envelope version, compiler version, and test-suite hash.
- I-08: Every runtime verdict returns control ID, version, reason codes, evidence references, and lineage pointer.
- I-09: A source change invalidates affected derived artifacts until impact analysis and required re-admission complete.
- I-10: An unavailable extraction model cannot alter already admitted controls.
- I-11: Hosted compilation is semantically reproducible from open-core artifacts and declared model/configuration inputs.

- I-12: Customer documents are never used for model training without explicit separate authorization; public Lab prohibits confidential content at launch.
- I-13: Parser and model stages operate without authority to publish or deploy controls.
- I-14: Publisher and reviewer powers are separable; revocation/correction authority is separately recorded.
- I-15: Correction invalidates unsafe reliance and preserves the historical record rather than overwriting it.

3. Prior Art, Commercial Landscape, and Build-Reuse Decisions

OIC MUST reuse mature foundations for commodity or well-standardized capabilities and MUST own the abstractions that define institutional meaning, review, portability, and correction. Architectural novelty is a hypothesis until differentiated capabilities are shown against open and commercial alternatives. The build decision is based on responsibility, semantic fit, failure isolation, licensing, offline deployment, version risk, replacement cost, and the ability to change a dependency without rewriting admitted institutional meaning.

3.1 Open standards, research, and open-source foundations

- **Docling**: strong open document conversion, layout, reading order, tables, and unified representation. It solves ingestion, not institutional authority or admission.
- **Akoma Ntoso / LegalDocML**: strong concepts for legal-document structure, identifiers, metadata, references, and amendment relationships. Full XML adoption would overburden the reference implementation and does not itself create an enterprise policy compiler.
- **ELI**: useful identifiers and metadata for legislation. It informs source identity and temporal relationships where applicable.
- **LegalRuleML**: strong normative vocabulary for obligations, permissions, prohibitions, temporal features, sources, defeasibility, and legal reasoning. It is an interchange foundation, not a user workflow for admitting inherited enterprise policy.
- **Better Rules and OECD Rules as Code**: strong process foundations for multidisciplinary rule modeling and official machine-consumable rules. They inform review methods but do not provide an enterprise-scale inherited-corpus compiler.
- **Catala**: strong literate executable statutory formalization and default/exception semantics, particularly for algorithmic law. It is not the generic first runtime target for broad enterprise policy.
- **OpenFisca**: mature operational infrastructure for tax and benefits calculation. It demonstrates shared computable rules but is domain-specific and expects human-authored formalization.
- **PROLEG and logic-programming traditions**: strong explainable inference after upstream formalization; they do not establish source authority or institutional admission.
- **DMN**: strong business-readable decision modeling for determinate decisions. It cannot be the canonical representation of authority, uncertainty, evidence duties, and non-decision provisions.
- **OPA/Rego**: mature domain-agnostic policy evaluation against structured data. It solves target enforcement after the rule exists.
- **Cedar**: a strong authorization-policy target with formal-analysis orientation. It is a later adapter, not the initial canonical representation.
- **W3C PROV-O, Web Annotation, SHACL, and RDF Dataset Canonicalization**: useful provenance, anchoring, validation, and canonicalization concepts. OIC uses compact JSON-native contracts first and maps to standards where interoperability justifies it.
- **ZTL**: project-controlled logical-warrant component intended to preserve T/F/Z, contradiction, dependency, and cannot-determine behavior after grounds are admitted. Its reuse status is provisional until the verification dossier in Section 3.4 is complete.
- **VEIP**: project-controlled execution-continuity component intended to preserve proposed, admitted, executed, recorded, relied-upon, and corrected states. Its reuse status is provisional until the verification dossier in Section 3.4 is complete.

3.2 Commercial and adjacent product landscape

This landscape is based on public vendor materials reviewed in July 2026. It is not a hands-on product evaluation, procurement recommendation, or assertion that a capability is absent from non-public roadmaps. “Public gap” means the reviewed materials did not establish the full OIC property under comparison.

Product/category	Publicly stated capability relevant to OIC	Relationship to OIC	Public gap OIC must test rather than assume
ServiceNow Policy and Compliance Management	Centralized policies, standards, internal controls, cross-mapping, workflow, assessment, and continuous monitoring	Major incumbent for policy/control lifecycle and enterprise workflow	Public materials do not establish open source-to-admitted-envelope compilation with target-independent runtime artifacts and field-level source lineage
IBM OpenPages Regulatory Compliance and Policy Management	Breaks regulations into requirement catalogs, maps requirements to controls/policies, manages change, tasks, workflows, and rules	Major GRC and regulatory-change alternative	Public materials do not establish automated inherited-document compilation into an open portable runtime envelope with explicit unknown/CANNOT semantics
CUBE RegPlatform	Automated regulatory intelligence, change analysis, and mapping to policies, controls, and teams	Strong regulatory-source and impact-analysis alternative	Primary public focus is regulatory intelligence and mapping; OIC must prove differentiated internal-corpus admission, executable projection, and correction lineage
LogicGate Risk Cloud	Policy/procedure lifecycle, obligation/control linking, evidence, exceptions, workflows, and regulatory integrations	Configurable GRC workflow alternative	Public materials do not establish an open canonical semantic compiler from source spans to portable runtime enforcement contracts
Norm Ai	Converts regulations and firm standards into legal/compliance AI agents and determinations with explanations	Directly adjacent legal/compliance automation competitor	Proprietary agent platform; OIC must prove open reproducibility, institution-owned envelope portability, admission records, and change-impact behavior
OneTrust AI Governance and Runtime Controls	AI inventory, policy governance, machine-readable rules, runtime monitoring, SDK controls, and	Direct runtime-governance competitor and important baseline for product positioning	OIC must not claim differentiation until it demonstrates inherited-corpus source anchoring, authority reconstruction,

Product/category	Publicly stated capability relevant to OIC	Relationship to OIC	Public gap OIC must test rather than assume
	enforcement		scoped admission, and open portable envelopes beyond public marketing descriptions
Credo AI Policy Center / Agent Governor	Policy packs, technical controls, governance workflows, and transformation of enterprise intent into runtime controls	Direct agent-governance competitor and important baseline	OIC must prove transparent source-to-control compilation, open schemas, independent local reproduction, and exact uncertainty/correction semantics
Rules and decision-engine vendors	Authoring and executing explicit decision logic or authorization policy	Likely target engines and integration partners rather than upstream compiler substitutes	They generally require structured rules; OIC must prove it can produce those rules safely from inherited institutional sources
Consulting-led policy-to-control transformation	Human experts interpret policy, design controls, implement workflows, and document evidence	Current manual incumbent and commercial baseline	OIC must show equal or better bounded quality with lower active time, stronger lineage, and faster change impact without claiming replacement of expert judgment

The competitive hypothesis is narrower than “no competitor does this.” It is:

Publicly available products solve substantial parts of regulatory intelligence, GRC workflow, policy management, AI governance, legal analysis, and runtime enforcement. OIC is designed to make the source-to-admitted-portable-control transformation open, inspectable, benchmarked, and independently reproducible. Whether this distinction is valuable and superior remains an empirical question.

3.3 Build-reuse matrix

Module	Decision	Foundation	Time value	Strategic control	Version risk	Rationale
M1 Corpus Intake Gateway	Reuse	Docling	High	Low	Medium	Parsing is commodity; wrap for security, hashes, anchors, and deterministic manifests.
M2 Source-Anchored Document Model	Borrow/rebuild	Akoma Ntoso, ELI, Web Annotation	High	Medium	Low	Use stable identity, hierarchy, references, amendments, selectors; implement compact JSON-native enterprise model.
M3 Corpus Authority Registry	Build	Legal source/authority concepts	Medium	High	Low	No accepted component has been identified that supplies OIC’s scoped inherited-source authority graph and admission boundary.
M4 Normative Clause Extractor	Borrow/rebuild	Constrained LLM extraction, LegalRuleML vocabulary	High	High	Medium	Use structured generation but rebuild schema, evidence binding, alternatives, confidence calibration, and no-silent-default behavior.
M5 Institutional IR	Build	LegalRuleML, DMN, provenance concepts	Medium	Very high	Low	Canonical target-independent representation is the core asset and must remain under OIC control.
M6 Dependency and Warrant Engine	Provisional project reuse	ZTL	Potentially high	High	High until dossier complete	Do not treat as accepted dependency until repository, license, interface, fixtures, conformance, and replacement boundary are pinned.
M7 Ambiguity and Conflict Docket	Build	Issue prioritization and graph-analysis concepts	Medium	Very high	Low	Novel minimization of blocking review questions across authority, definitions, evidence, logic, and change.
M8 Institutional Review Console	Borrow/rebuild	Better Rules, DMN UX	High	High	Low	Borrow practitioner-readable patterns; rebuild admission workflow, authority checks, impact display, and disposition record.
M9 Open Control Envelope Generator	Build	Policy schemas and authorization contracts	Medium	Very high	Low	Portable institution-owned boundary is the central differentiating abstraction and must be independent of target engines.
M10 Multi-Target Compiler	Reuse	OPA/Rego; DMN; later Cedar	High	Medium	Medium	Use mature runtimes; own target adapters and equivalence tests.
M11 Test and Equivalence Harness	Borrow/rebuild	OPA tests, mutation testing, SHACL	High	Very high	Low	Existing test tools do not prove source-to-control, ambiguity, authority, and cross-target preservation.

Module	Decision	Foundation	Time value	Strategic control	Version risk	Rationale
M12 Runtime and Evidence Adapter	Provisional project reuse	VEIP	Potentially high	High	High until dossier complete	Do not treat as accepted dependency until specification, license, interface, lifecycle fixtures, conformance, and replacement boundary are pinned.

3.4 Component count and novelty hypothesis

The baseline architecture contains twelve modules:

- **2 accepted substantial reuses:** M1 and M10.
- **2 provisional project-controlled reuses:** M6 and M12.
- **4 borrowed but rebuilt behind OIC-owned contracts:** M2, M4, M8, and M11.
- **4 built from scratch:** M3, M5, M7, and M9.

If the ZTL and VEIP verification dossiers pass, the practical count becomes four substantially reused, four borrowed/rebuilt, and four built from scratch. If either dossier fails, the adapter contract remains but the implementation decision must be reopened through an ADR.

The novelty hypothesis is the integrated transformation and guarantees, not any single parser, model, graph, or policy engine. The hypothesized differentiator is the combination of authority-aware corpus construction, target-independent admitted meaning, minimum blocking review, portable control envelopes, deterministic warrant, runtime evidence, and correction lineage. The claim remains unproven until comparative evaluation and independent consumption occur.

3.5 Dependency acceptance criteria

Every external or project-controlled dependency requires:

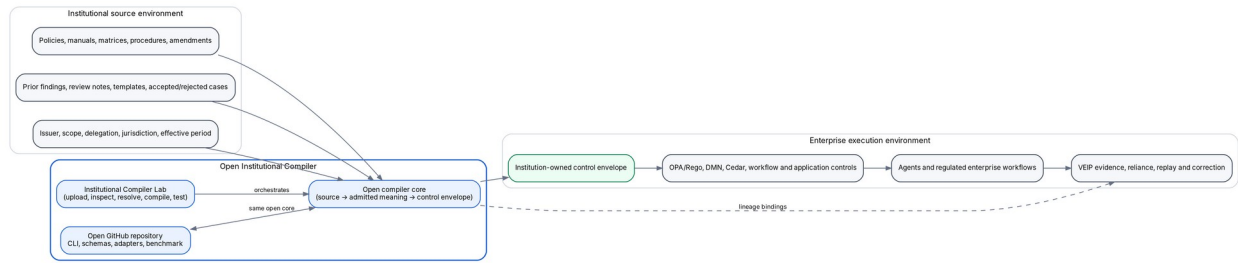
- exact version, commit, or content digest pinned in the release lockfile;
- owner, maintainer, license, support status, and repository or artifact location;
- adapter contract isolating upstream schemas and APIs from canonical OIC entities;
- offline/on-prem behavior and hosted dependencies documented;
- known parsing, semantic, security, and performance limitations;
- replacement strategy, failure behavior, and compatibility tests;
- SBOM and vulnerability-monitoring coverage;
- evidence that upstream updates cannot silently alter admitted output or benchmark results.

For **ZTL and VEIP**, release freeze additionally requires a verification dossier containing:

1. immutable source/specification reference and cryptographic digest;
2. named owner and maintenance authority;
3. license and permitted open/hosted use;
4. interface schemas and semantic boundary statement;
5. published fixtures and conformance tests;
6. reproducible build or independently verifiable binary provenance;
7. known limitations and claim ceiling;
8. failure and fallback behavior;
9. replacement architecture if the component is unavailable;
10. independent reproduction status explicitly labeled.

Until these dossiers pass, M6 and M12 may be implemented as provisional adapters, but their absence **MUST NOT** be concealed and their outputs **MUST NOT** support public maturity claims.

4. System Architecture and Responsibility Boundaries



System context: two artifacts share one open semantic core.

4.1 Four planes



Compilation, admission, enforcement, and continuity planes.

Compilation plane

Transforms immutable source artifacts into a source-anchored graph, authority records, candidate normative units, interpretation alternatives, and blocking questions. It is allowed to propose meaning but not to admit or publish it.

Admission plane

Presents source, proposed operational meaning, alternatives, examples, affected controls, and unresolved issues to authorized reviewers. It records scoped dispositions and creates immutable admission records.

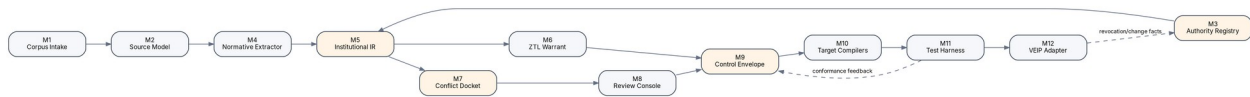
Enforcement plane

Produces Open Control Envelopes and target-specific artifacts only from admitted records. It validates target equivalence and serves runtime decisions through target engines or embedded evaluators.

Continuity plane

Preserves logical warrant, exact versions, evidence, execution results, reliance, revocation, change impact, historical replay, and correction. It prevents a correct historical record from being overwritten by a later interpretation.

4.2 Module pipeline



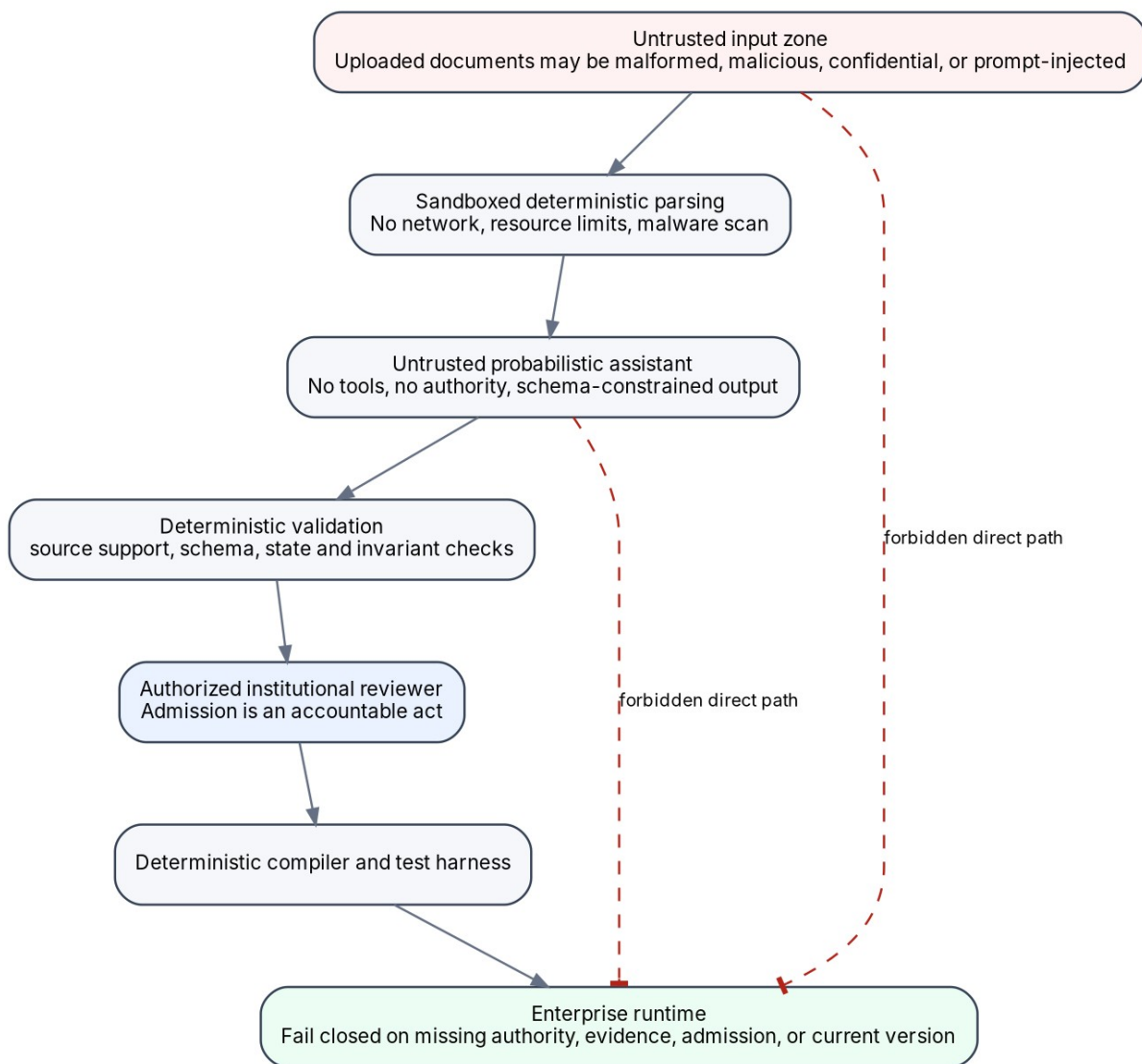
Twelve-module compiler pipeline.

4.3 Chosen implementation topology

The seven-day reference implementation will be a **modular monolith** with explicit Python package boundaries, typed contracts, and durable artifacts. This is a deliberate tradeoff: it minimizes distributed-systems overhead while preserving boundaries that can later become services. The hosted Lab adds a web UI, authentication, tenant isolation, object storage, database, and job orchestration around the same compiler packages.

No module may directly mutate another module's persisted artifact. Each stage produces a versioned, hashed output. Reprocessing creates a new version and an explicit lineage edge.

4.4 Trust boundaries



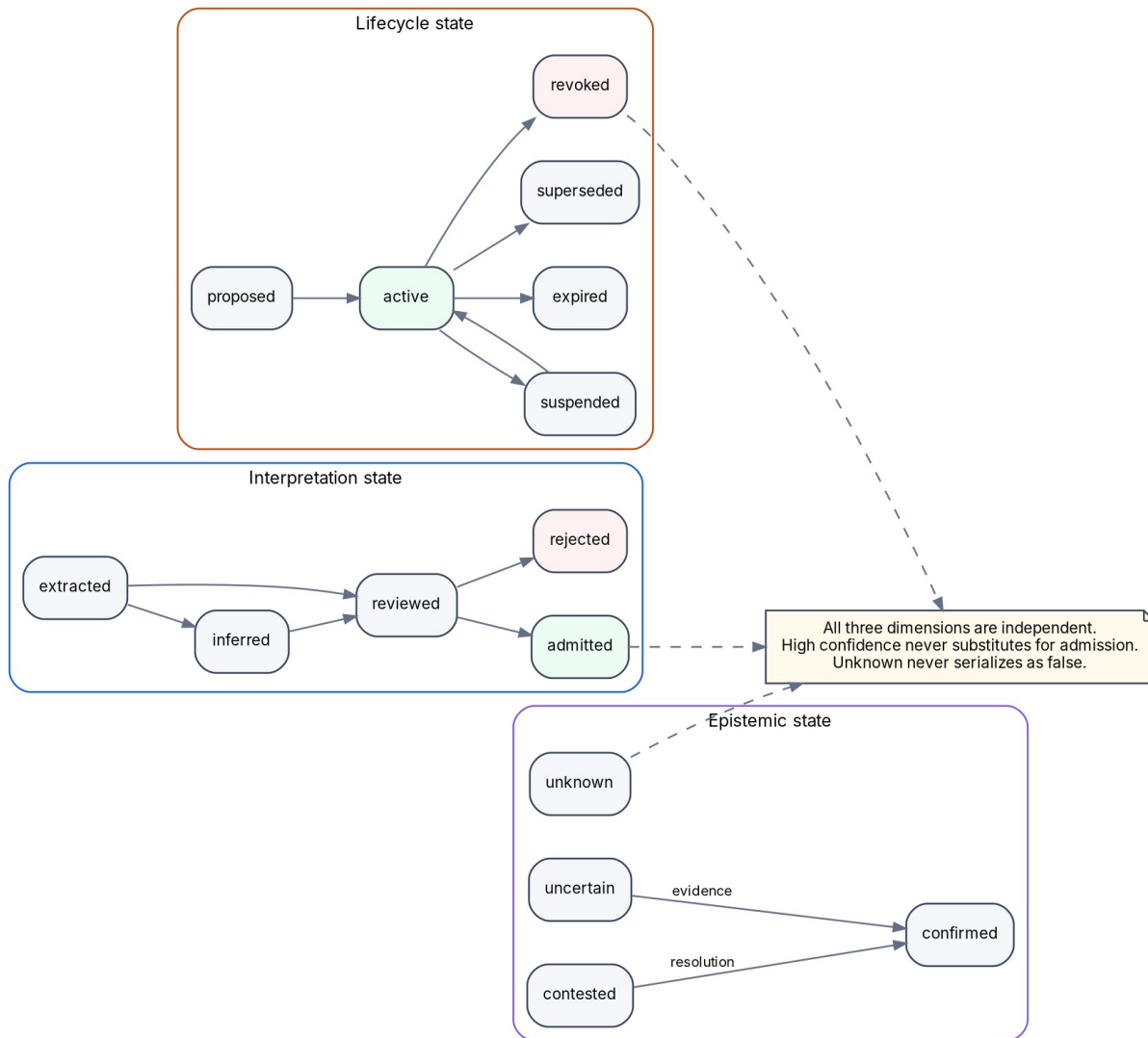
Primary trust boundaries and untrusted inputs.

The following are untrusted by default:

- uploaded files and embedded active content;
- OCR and parser output;
- model prompts and generated structures;
- external metadata supplied by users;
- runtime facts supplied by calling systems;
- target compiler output until conformance tests pass;
- community policy packs and adapters;
- benchmark submissions from external parties.

Authority and admission cannot be delegated to a model or parser process. Publishing requires a distinct authorized action and a valid admission set.

4.5 Semantic state model



Independent interpretation, epistemic, lifecycle, logical, and operational states.

State dimensions are deliberately independent. A proposition can be institutionally admitted yet epistemically contested. An authority can be semantically clear yet expired. A logical formula can evaluate to T while the operational verdict is DENY because the actor lacks current authority or evidence. The model MUST NOT collapse these dimensions into one confidence score.

4.6 ZTL and VEIP boundaries

ZTL provisional boundary

ZTL is a project-controlled candidate dependency. OIC may project admitted propositions and grounding facts to a pinned ZTL interface and consume deterministic T/F/Z, dependency, contradiction, and CANNOT artifacts. ZTL MUST NOT validate source authenticity, determine authority, interpret prose, or create institutional admission. Until its verification dossier is complete, the adapter is experimental and a deterministic OIC reference evaluator or review-only fallback MUST remain available.

VEIP provisional boundary

VEIP is a project-controlled candidate dependency. OIC may bind a proposed action to the exact envelope, authority state, facts, warrant, evidence, execution result, reliance, and correction record through a pinned VEIP interface. VEIP **MUST NOT** originate policy meaning or authority. Until its verification dossier is complete, the adapter is experimental and the reference implementation **MUST** be able to emit a transparent lifecycle/evidence contract without requiring an opaque service.

Separation invariant

Neither project-controlled component may be cited as evidence of OIC maturity merely because it is named in the architecture. Each must earn dependency acceptance through the same pinning, licensing, conformance, security, and replacement discipline applied to public dependencies.

5. End-to-End Reference Flow

The release demonstration MUST use a difficult but bounded procurement corpus rather than a toy two-page policy. The corpus will contain a primary policy, definitions annex, approval matrix, later amendment, conflicting procedure, expired delegation, emergency exception, evidence requirements, an open-textured clause, and transaction cases.

5.1 Golden path

1. **Upload and register.** Files are malware-scanned, content-hashed, assigned stable IDs, and placed in an immutable project manifest.
2. **Parse and anchor.** Layout, hierarchy, tables, cells, footnotes, and cross-references are recovered. Every node receives page/region/character selectors and content hashes.
3. **Classify authority.** A steward records issuer, type, authority basis, organizational scope, subject scope, effective period, binding strength, supersession, and review owner.
4. **Extract candidates.** The model proposes actors, actions, modalities, conditions, exceptions, evidence duties, delegations, definitions, escalation, review, remedies, and temporal triggers with exact source support and alternatives.
5. **Resolve structure.** Definitions and cross-references are connected; candidate dependencies and conflicts are computed.
6. **Generate docket.** The system ranks the minimum unresolved questions that block the greatest number or highest consequence of controls.
7. **Review.** A practitioner sees source, proposed meaning, alternatives, examples, affected controls, and current limitations. The practitioner admits, qualifies, modifies, rejects, escalates, or marks non-automatable.
8. **Create canonical IR.** Only admitted records become eligible for enforcement projections.
9. **Generate warrant.** OIC projects admitted propositions and facts to ZTL, preserving T/F/Z and explicit missing grounds.
10. **Build envelope.** OIC creates target-independent control contracts with actor, action, authority chain, conditions, evidence, exceptions, decision behavior, effective period, and correction hooks.
11. **Compile and test.** The envelope becomes Rego and JSON artifacts; generated tests and invariants run.
12. **Evaluate runtime request.** A transaction returns ALLOW, DENY, CANNOT, ESCALATE, or NOT_APPLICABLE with reason codes and lineage.
13. **Record and rely.** VEIP binds the action, envelope, warrant, evidence, runtime result, and execution outcome.
14. **Change source.** An amendment or revocation creates a new source version. The dependency graph identifies affected interpretations, controls, tests, and reliance.
15. **Invalidate and correct.** Unsafe artifacts are suspended until re-review. Historical decisions remain reconstructable and affected reliance can be reviewed or corrected.

5.2 Representative runtime response

```
{
  "evaluation_id": "eval_01JZ...",
  "verdict": "CANNOT",
  "control_id": "PROC-APPROVAL-017",
  "control_version": "1.3.0",
  "reason_codes": ["UNRESOLVED_DELEGATION_SCOPE"],
  "missing_inputs": ["regional_director_limit"],
  "source_anchors": [
    "src:procurement-policy-2026#p14-clause4.3",
    "src:delegation-matrix-2026#Limits!E12"
  ],
  "warrant_id": "ztl_warrant...",
  "evidence_pack_id": "aep...",
  "lineage_uri": "/lineage/eval_01JZ..."
}
```

5.3 Failure paths

Failure	Required behavior	Prohibited behavior
Unreadable or malicious document	Quarantine file; retain manifest; report parser/security error	Partial semantic output treated as complete
Incomplete table extraction	Mark affected cells/clauses uncertain; create review task	Infer missing thresholds from neighboring values
Contradictory definitions	Create conflict object and block dependent controls	Choose the most recent-looking definition without authority analysis
Missing delegation	CANNOT or ESCALATE; expose required	Assume role title implies authority

Failure	Required behavior	Prohibited behavior
	authority evidence	
Expired source	Exclude from active authority except historical replay	Allow based on stale control
Prompt injection in document	Treat text as content only; no tools/network; flag malicious patterns	Follow embedded instructions
Model timeout or invalid output	Retry boundedly; preserve stage state; no publication	Fallback to free-form output
ZTL unavailable	Compilation may continue to review-only; publication requiring warrant blocked	Invent warrant or convert unknown to false
Target compiler mismatch	Fail artifact conformance; do not publish target	Publish because source IR appears valid
Runtime facts unavailable	CANNOT, DENY, or ESCALATE according to admitted policy	Default missing evidence to satisfied
Hosted outage	Local/open artifacts remain usable; published envelopes continue under declared cache policy	Require hosted service for every decision unless customer chose that mode

6. Detailed Module Designs

6.1 M1 - Corpus Intake Gateway

Attribute	Design
Build decision	Substantially reuse
Foundation	Docling plus sandboxed file-handling services
Technical problem	Enterprise corpora contain PDFs, DOCX, XLSX, HTML, images, annexes, tables, and scans. Semantic quality is impossible without stable layout, reading order, table cells, page coordinates, and immutable source identity.
Performance/capacity	Reference target: 50 digital pages in <=5 minutes p95; 50 scanned pages in <=15 minutes p95. Public Lab limit: 100 pages/project and 50 MB/file.

Responsibility

- Validate upload type, size, malware result, and project limits.
- Compute cryptographic hashes before transformation.
- Convert supported formats into a unified structural representation.
- Preserve pages, blocks, headings, lists, tables, cells, images, footnotes, and reading order.
- Emit deterministic ingestion manifests and parser diagnostics.

Interface contract

Inputs

- Original file bytes
- User-supplied source metadata
- Tenant/project policy
- Parser configuration

Outputs

- SourceDocument
- ParsedDocumentArtifact
- IngestionManifest
- ParserDiagnostic

Guarantees

- Original bytes remain immutable and addressable.
- Every parsed node links to the original file and parser version.
- Parser uncertainty and extraction gaps are explicit.
- No semantic authority or normative type is assigned.

Explicit non-responsibilities

- Determine which document is authoritative.
- Interpret clauses or infer missing text.
- Execute embedded macros, links, scripts, or document instructions.

Failure modes

Failure	Required response
Malware or active content	Quarantine and block parsing.
Unsupported format	Reject with conversion guidance.
OCR confidence below threshold	Mark regions uncertain; request human correction.
Parser crash	Retry in isolated worker; preserve immutable source.
Zip bomb/path traversal	Reject before extraction.

Security controls

- Sandbox parsers with no outbound network and read-only input mount.
- Enforce decompressed-size, page-count, recursion, CPU, and memory limits.
- Strip active content from derivative views while preserving original hash.

Acceptance tests

- A 50-page digital PDF produces stable page/block/table anchors.
- A spreadsheet cell is addressable by workbook, sheet, row, column, and hash.
- Re-running the pinned parser on the same bytes/config produces an equivalent manifest.
- A malicious fixture cannot reach network, filesystem outside sandbox, or control plane.

6.2 M2 - Source-Anchored Document Model

Attribute	Design
Build decision	Borrow concepts and rebuild
Foundation	Akoma Ntoso, ELI, W3C Web Annotation, PROV-O
Technical problem	Raw parser output is not a stable institutional source model. Controls need durable identifiers across page layout changes, cross-references, amendments, and heterogeneous document types.
Performance/capacity	Anchor lookup p95 <=200 ms for a 5,000-page enterprise corpus using project-scoped indexes.

Responsibility

- Normalize parser output into compact JSON-native source nodes.
- Assign stable document, section, clause, table, cell, definition, footnote, and annex identifiers.
- Record exact selectors: page, bounding box, character offsets, cell coordinates, and content hash.
- Represent references, amendments, supersession, versions, and manifestations.
- Support source rendering and bidirectional navigation.

Interface contract

Inputs

- ParsedDocumentArtifact
- Original file hash
- Declared source metadata

Outputs

- AnchoredSourceGraph
- SourceAnchorIndex
- SourceRenderManifest

Guarantees

- Every downstream proposition can cite one or more precise source anchors.
- Source node identity is distinct from a particular rendered manifestation.
- Changes create new versions and explicit relationships rather than overwriting history.

Explicit non-responsibilities

- Adopt full Akoma Ntoso XML as internal storage.
- Resolve authority hierarchy.
- Claim semantic equivalence from structural similarity.

Failure modes

Failure	Required response
Unstable OCR text	Use region/image anchors and mark textual selector uncertain.
Broken cross-reference	Create unresolved reference rather than guessing.
Table merge ambiguity	Retain original cell geometry and create parser diagnostic.
Duplicate clauses	Preserve distinct source identities and expose similarity.

Security controls

- Escaped rendering prevents active HTML/script execution.
- Anchor APIs enforce tenant and project authorization.
- Original bytes and derivative text have separate access classifications.

Acceptance tests

- Every gold normative clause in the reference corpus is addressable.
- Source-anchor precision/recall meets frozen benchmark thresholds.
- Amended and superseded nodes remain replayable.
- Lineage UI opens the exact source region from a control field.

6.3 M3 - Corpus Authority Registry

Attribute	Design
Build decision	Build from scratch
Foundation	Legal source, delegation, provenance, and temporal concepts
Technical problem	A parsed document says something; it does not establish why the document controls a person, action, workflow, jurisdiction, or time. Enterprises need an explicit authority ecology rather than a folder of apparently equal files.
Performance/capacity	Incremental precedence recomputation <=5 seconds for 10,000 authority edges.

Responsibility

- Record issuer, source type, authority basis, jurisdiction, organizational scope, subject scope, binding strength, effective period, and review owner.
- Represent superiority, delegation, subdelegation, implementation, interpretation, amendment, supersession, conflict, and customary-practice relations.
- Calculate candidate precedence and flag missing or circular authority.
- Separate declared metadata from verified evidence and reviewer-admitted status.
- Expose historical authority state for replay.

Interface contract

Inputs

- AnchoredSourceGraph
- User/steward metadata
- Organizational role directory
- External authority evidence where available

Outputs

- AuthorityRecord
- AuthorityRelationship
- AuthorityPrecedenceGraph
- AuthorityIssue

Guarantees

- No source is treated as authoritative merely because it is recent, titled policy, or supplied by an administrator.
- Authority status is scoped and time-bound.
- Delegation chains retain origin, constraints, and subdelegation status.

Explicit non-responsibilities

- Determine ultimate legal validity.
- Resolve conflicts without an admitted precedence rule or steward disposition.
- Infer delegation from job title or common practice alone.

Failure modes

Failure	Required response
Missing authority basis	Mark ungrounded; block dependent publication.
Circular delegation	Create contradiction and escalation.
Conflicting superior sources	Create authority conflict docket item.
Expired evidence	Suspend dependent authority state.
Directory mismatch	Require steward resolution.

Security controls

- Authority Steward is a privileged role separate from Compiler Operator.
- Changes are append-only and signed.
- Bulk import cannot auto-admit authority records.

Acceptance tests

- Gold authority and delegation relations meet benchmark F1.
- Expired and superseded sources are detected with ≥ 0.95 recall.
- A forged metadata update cannot alter an admitted graph without authorized disposition.
- Historical replay reconstructs authority as of a requested timestamp.

6.4 M4 - Normative Clause Extractor

Attribute	Design
Build decision	Borrow capabilities and rebuild
Foundation	Constrained LLM extraction, LegalRuleML vocabulary, rule-extraction

Attribute	Design
	literature
Technical problem	Institutional prose must be decomposed into candidate actors, actions, modalities, conditions, exceptions, evidence, discretion, review, remedies, and temporal behavior without pretending that extraction is admission.
Performance/capacity	Reference target: <=15 minutes p95 for extraction of a 50-page corpus, excluding human review.

Responsibility

- Detect potentially normative and definitional source units.
- Generate typed candidate units using a versioned JSON schema.
- Bind every field to exact evidence spans or mark it derived/inferred.
- Produce alternative interpretations and calibrated confidence.
- Classify non-automatable and open-textured provisions.
- Run schema validation, self-consistency checks, and model-independent postprocessors.

Interface contract

Inputs

- AnchoredSourceGraph
- Authority context
- Extraction profile
- Model/provider configuration

Outputs

- CandidateNormativeUnit
- InterpretationAlternative
- ExtractionDiagnostic
- NonNormativeClassification

Guarantees

- Model output is never directly enforceable.
- No silent defaults for missing actor, modality, exception, time, or evidence.
- Quoted evidence remains below configured length and links to anchors.
- Model/provider and prompt versions are recorded.

Explicit non-responsibilities

- Admit interpretation.
- Resolve policy conflicts.
- Claim high confidence equals institutional correctness.
- Use external tools or network based on document content.

Failure modes

Failure	Required response
Invalid schema	Repair once under bounded procedure, then fail visibly.
Unsupported field	Reject unit or preserve extension namespace.
Hallucinated source	Fail source-support validator.
Contradictory alternatives	Preserve alternatives and docket item.

Failure	Required response
Model unavailable	Pause extraction; preserve all prior artifacts.

Security controls

- Model gateway strips tool access and outbound browsing.
- Prompt-injection detectors label hostile instructions but do not treat detection as sole defense.
- Customer content is not retained by provider beyond approved terms and is never used for training without authorization.

Acceptance tests

- Normative Field macro-F1 ≥ 0.85 on frozen test corpus.
- Unsupported executable-field precursor rate is measured and all unsupported fields remain non-executable.
- Ambiguous clauses generate alternatives or uncertainty labels.
- Changing model cannot change admitted artifacts without new version/review.

6.5 M5 - Typed Institutional Intermediate Representation

Attribute	Design
Build decision	Build from scratch
Foundation	LegalRuleML, provenance, decision modeling, compiler IR design
Technical problem	Direct compilation from prose into Rego, DMN, or code ties institutional meaning to one target and forces reinterpretation for each new runtime. OIC needs one canonical representation of admitted meaning, uncertainty, authority, and lifecycle.
Performance/capacity	Validation and canonicalization ≤ 10 seconds for 10,000 IR nodes in pilot scale.

Responsibility

- Represent sources, definitions, actors, roles, authority, delegation, actions, resources, modalities, conditions, evidence, exceptions, discretion, escalation, review, remedies, time, jurisdiction, and unresolved questions.
- Maintain independent interpretation, epistemic, and lifecycle states.
- Version nodes and edges semantically.
- Validate graph constraints and extension profiles.
- Provide canonical serialization and deterministic hashing.

Interface contract

Inputs

- CandidateNormativeUnit
- Authority records
- Review dispositions
- Source anchors

Outputs

- InstitutionalIRNode
- InstitutionalIREdge
- IRSnapshot
- IRValidationReport

Guarantees

- Target code is never canonical meaning.
- Extracted/inferred items cannot masquerade as admitted.
- Unknown, contested, suspended, expired, and revoked states survive serialization.
- Canonical hashes change when semantically material fields change.

Explicit non-responsibilities

- Require RDF storage in the reference release.
- Claim a universal ontology.
- Collapse discretionary procedure into deterministic outcome.

Failure modes

Failure	Required response
Schema incompatibility	Reject import with migration guidance.
Dangling source or authority edge	Fail publication readiness.
Invalid state transition	Reject and audit.
Hash mismatch	Quarantine artifact.
Extension collision	Require namespaced profile and conformance rule.

Security controls

- Canonical snapshots are signed at admission and publication boundaries.
- Mutating admitted records requires new versions, never in-place edits.
- Deserializer rejects unknown executable semantics by default.

Acceptance tests

- Round-trip serialization preserves all gold fields and states.
- Canonicalization is deterministic under the pinned implementation.
- Every enforceable node has source, authority, and admission references.
- Schema migrations have fixture-based backward compatibility tests.

6.6 M6 - Dependency and Warrant Engine

Attribute	Design
Build decision	Provisional project-controlled reuse; verification dossier required
Foundation	ZTL project artifacts through a pinned adapter
Technical problem	After grounds are admitted, the system must calculate which conclusions follow, which facts are missing, how dependencies propagate, and when a result is contradiction or CANNOT without collapsing unknown into false.
Performance/capacity	Warrant evaluation p95 ≤ 10 ms for reference formulas; complex graphs have explicit ceilings.

Responsibility

- Project admitted IR propositions into ZTL-compatible atoms and formulae.
- Compute T/F/Z outcomes and dependency graphs.
- Emit recomputable warrant artifacts and missing-ground lists.
- Reconcile seams with SEWN, CONTRADICTION, or CANNOT behavior.
- Support expiry, revoke, and bounded refinement events.

Interface contract

Inputs

- Admitted IR snapshot
- Grounding facts
- Temporal context
- ZTL profile/version

Outputs

- LogicalDependencyGraph
- WarrantArtifact
- MissingGround
- ContradictionRecord

Guarantees

- Verdict is not represented as world state.
- F is not described as grounded negative fact unless grounds establish it.
- Z never serializes as F.
- Logical proof is bound to exact inputs, formula, and engine version.

Explicit non-responsibilities

- Validate whether a source is authentic or authoritative.
- Interpret prose.
- Determine operational ALLOW/DENY alone.
- Resolve contested institutional facts.

Failure modes

Failure	Required response
Unsupported formula	Block warrant-required publication or route to human judgment.
Engine unavailable	Review-only mode; no fabricated warrant.
Contradiction	Return contradiction and affected dependency set.
Ground expired	Recompute and propagate invalidation.
Version mismatch	Reject replay until byte-identical compatible engine/profile is available.

Security controls

- Run deterministic engine in isolated process.
- Pin binaries/commits and verify hashes.
- Grounding facts are authorized and classified separately from source policy.

Acceptance tests

- Dependency dossier passes the requirements in Section 3.5 before release freeze.
- Published fixtures reproduce exact verdict and warrant hash.
- Unknown preservation invariant passes all mutation tests.
- Dependency invalidation identifies all affected conclusions.
- ZTL boundary tests prove it cannot self-certify source authority.

6.7 M7 - Ambiguity and Conflict Docket

Attribute	Design
Build decision	Build from scratch
Foundation	Graph impact analysis, issue minimization, human-in-the-loop review
Technical problem	Showing reviewers hundreds of extracted clauses does not solve the operational bottleneck. The system must identify the smallest set of questions whose resolution unlocks the largest or highest-risk portion of the control graph.
Performance/capacity	Docket generation ≤ 10 seconds after graph availability for 1,000 candidate units.

Responsibility

- Detect undefined terms, competing definitions, missing actors, missing delegation, conflicting thresholds, exception ambiguity, temporal conflict, evidence gaps, open-textured standards, and ZTL unknowns.
- Group duplicate or causally related issues.
- Calculate blocking impact across candidate controls and consequences.
- Prioritize by consequence, fan-out, uncertainty, and reviewer role.
- Generate source-grounded, answerable review questions.

Interface contract

Inputs

- Candidate units
- Authority graph
- IR draft
- Cross-reference graph
- Warrant diagnostics

Outputs

- UnresolvedQuestion
- ReviewDocket
- BlockingSet
- ImpactSummary

Guarantees

- No issue is auto-resolved solely to maximize compilation rate.
- Question text links to source and affected controls.
- Priority score is explainable and does not equal substantive correctness.
- Resolved questions retain disposition and historical alternatives.

Explicit non-responsibilities

- Make legal judgment.
- Use model confidence as the sole priority.
- Hide low-impact uncertainty from downloadable artifacts.

Failure modes

Failure	Required response
Issue explosion	Cluster with reversible membership; never delete underlying issues.
False conflict	Allow reviewer dismissal with reason; feed benchmark.

Failure	Required response
Missed blocking issue	Detected by mutation/held-out evaluation and incident process.
Wrong reviewer routing	Escalate and preserve audit trail.

Security controls

- Question generation cannot publish controls.
- Sensitive source excerpts follow least-privilege access.
- Priority manipulation is logged and benchmarked.

Acceptance tests

- Ambiguity recall ≥ 0.85 ; false-resolution rate ≤ 0.05 .
- Unknown-to-false conversion equals zero.
- Docket reduces review volume while preserving all gold blocking issues.
- A ten-minute orientation allows practitioners to understand impact and disposition options.

6.8 M8 - Institutional Review Console

Attribute	Design
Build decision	Borrow patterns and rebuild
Foundation	Better Rules, DMN practitioner views, evidence-review UX
Technical problem	Institutional experts understand source language, examples, and operational effects. They should not need to learn Rego, graph query languages, or formal logic to control the compilation.
Performance/capacity	Review card load p95 ≤ 2 seconds; lineage expansion p95 ≤ 500 ms for reference corpus.

Responsibility

- Show source text and exact location beside proposed meaning.
- Display actor, authority, conditions, exceptions, evidence, examples, alternatives, and affected controls.
- Support admit, admit with qualification, modify, reject, escalate, and non-automatable dispositions.
- Verify reviewer role and admission authority.
- Create immutable AdmissionRecord and approval history.
- Support collaborative comments without confusing discussion with disposition.

Interface contract

Inputs

- ReviewDocket
- Candidate units
- Authority context
- Generated examples/tests
- Reviewer identity and role

Outputs

- ReviewDisposition
- AdmissionRecord
- ReviewComment
- EscalationTask

Guarantees

- Only authorized roles can create scoped admission.
- Reviewer sees material downstream impact before disposition.
- Admission records bind exact source, interpretation, scope, and versions.
- UI labels candidate, admitted, qualified, rejected, and non-automatable states distinctly.

Explicit non-responsibilities

- Describe candidates as approved before admission.
- Let platform administrators silently act as policy reviewers.
- Hide machine confidence or known limitations.
- Require code editing for ordinary review.

Failure modes

Failure	Required response
Concurrent edits	Optimistic locking and explicit merge/re-review.
Reviewer authority revoked	Prevent new admission and flag affected pending tasks.
Source changes mid-review	Invalidate card and require refreshed review.
UI outage	Export review pack; no loss of draft/disposition state.
Unauthorized action	Deny and audit.

Security controls

- RBAC/ABAC with separation of reviewer, publisher, and correction roles.
- Strong authentication; enterprise SSO in pilot.
- Admission actions are signed or integrity-bound and append-only.

Acceptance tests

- Orientation ≤ 15 minutes.
- Median review time ≤ 3 minutes per candidate item on usability corpus.
- Reviewer agreement and task completion are measured.
- Every admitted record can be reconstructed without UI state.

6.9 M9 - Open Control Envelope Generator

Attribute	Design
Build decision	Build from scratch
Foundation	Authorization contracts, policy bundles, typed enforcement schemas
Technical problem	A runtime cannot enforce a legal essay or generic graph. It needs a bounded portable contract that states who may do what, under which authority and conditions, with which evidence, exceptions, time limits, and failure behavior.
Performance/capacity	Generate and validate 1,000 envelopes in ≤ 15 seconds on reference hardware.

Responsibility

- Transform admitted IR into portable target-independent control envelopes.
- Distinguish deterministic enforcement, human judgment procedure, evidence obligation, escalation-only rule, and advisory content.
- Bind authority chain, required facts, evidence schema, decision behavior, effective period, revocation sources, and correction hooks.

- Assign stable IDs and semantic versions.
- Emit human-readable and machine-readable views.

Interface contract

Inputs

- Admitted IR snapshot
- Authority graph
- Warrant profile
- Compilation profile

Outputs

- OpenControlEnvelope
- EnvelopeManifest
- HumanControlMatrix

Guarantees

- Every executable field has source/admission support.
- Envelope declares on_missing, on_unknown, on_conflict, and on_error behavior.
- Envelope is independent of Rego, DMN, Cedar, or a vendor runtime.
- Envelope publication state is distinct from generation state.

Explicit non-responsibilities

- Create new meaning to satisfy a target.
- Convert every discretionary clause into an allow/deny rule.
- Omit limitations for convenience.

Failure modes

Failure	Required response
Missing runtime-observable input	Generate non-deployable envelope issue or escalation procedure.
Unsupported semantic type	Preserve non-executable obligation; block target projection if required.
Conflicting admitted rules	Apply admitted priority if present; otherwise CANNOT.
Stale dependency	Suspend envelope publication.

Security controls

- Envelope signing and digest verification.
- Publisher role separate from reviewer.
- Sensitive evidence schemas may be referenced without embedding source content.

Acceptance tests

- Source-supported executable fields ≥ 0.99 .
- Unsupported executable-field rate ≤ 0.01 .
- Human and JSON views are semantically equivalent under conformance tests.
- Envelope can compile to at least Rego and JSON runtime contract without reinterpretation.

6.10 M10 - Multi-Target Compiler

Attribute	Design
Build decision	Substantially reuse target engines

Attribute	Design
Foundation	OPA/Rego first; DMN export; Cedar later
Technical problem	Institutions use multiple runtimes. OIC must project one admitted envelope into target-specific artifacts without allowing target limitations or conveniences to mutate canonical meaning.
Performance/capacity	Compile 1,000 controls to a target in ≤ 20 seconds; runtime evaluation p95 ≤ 25 ms for local OPA reference.

Responsibility

- Compile supported envelope subsets to Rego and JSON runtime contracts.
- Export suitable decision logic to DMN.
- Maintain target capability profiles and unsupported-feature diagnostics.
- Generate target tests and source maps.
- Validate behavior against canonical test vectors.

Interface contract

Inputs

- OpenControlEnvelope
- Target profile
- Compiler configuration
- Canonical tests

Outputs

- TargetArtifact
- TargetSourceMap
- TargetDiagnostic
- CompilationManifest

Guarantees

- Target artifact is a derived projection, never the source of truth.
- Compiler records envelope, adapter, engine, and toolchain versions.
- Unsupported semantics fail visibly or remain non-executable.
- Published artifacts pass target and cross-target tests.

Explicit non-responsibilities

- Rewrite the envelope to improve target compile rate.
- Treat DMN or Rego as universal representations.
- Permit target-specific extension to bypass canonical admission.

Failure modes

Failure	Required response
Compile error	Fail publication; preserve diagnostics.
Behavior mismatch	Quarantine adapter version and block release.
Engine version drift	Pin or re-run full conformance before update.
Unsupported target feature	Return explicit capability gap.
Runtime bundle delivery failure	Retain prior approved bundle or fail closed per deployment policy.

Security controls

- Sandbox compiler toolchain.

- Sign artifacts and publish SBOM/provenance.
- No arbitrary code execution from envelope fields.

Acceptance tests

- OPA artifact compile rate and behavioral conformance meet frozen thresholds.
- Cross-target equivalence ≥ 0.98 once two executable targets exist.
- All target artifacts include source maps and manifests.
- Adapter replacement does not require IR migration.

6.11 M11 - Test and Equivalence Harness

Attribute	Design
Build decision	Borrow test infrastructure and rebuild semantics
Foundation	OPA tests, mutation testing, property testing, SHACL-style validation
Technical problem	Successful code generation does not prove preservation of source meaning, authority, uncertainty, exceptions, or change behavior. OIC needs tests across the entire transformation.
Performance/capacity	Full public benchmark completes within 60 minutes on documented reference hardware; fast PR suite ≤ 15 minutes.

Responsibility

- Generate positive, negative, boundary, exception, missing-evidence, unknown, conflict, expiry, revocation, and unauthorized-actor cases.
- Run source-to-field support checks.
- Run semantic mutation tests: must/may, and/or, actor swap, threshold, negation, time shift, exception toggle, priority change.
- Validate invariants and target behavior.
- Compare outputs across targets and compiler versions.
- Produce conformance, coverage, counterexample, and limitation reports.

Interface contract

Inputs

- Corpus gold annotations
- IR snapshot
- Control envelopes
- Target artifacts
- Runtime test vectors

Outputs

- ConformanceReport
- CoverageReport
- MutationReport
- CounterexampleSet
- ReleaseGateResult

Guarantees

- No release claim from unpinned corpus or post-hoc threshold.
- A passing target test does not erase source or authority failures.
- Failures retain reproducible inputs and hashes.

- Benchmark and customer test results are separated.

Explicit non-responsibilities

- Use model self-evaluation as sole quality evidence.
- Optimize against private held-out annotations.
- Hide regression by changing corpus or metric definition.

Failure modes

Failure	Required response
Flaky test	Quarantine and investigate; not counted as pass.
Corpus drift	Hash mismatch blocks comparison.
Baseline unavailable	No comparative claim.
Mutation survivor	Record semantic weakness and block relevant claim/release gate.
Metric implementation defect	Invalidate affected reports and reissue.

Security controls

- Benchmark gold data separated by access and role.
- External submissions run in sandbox.
- Results and thresholds signed at release freeze.

Acceptance tests

- All 15 invariants have executable tests.
- Behavioral conformance ≥ 0.95 on admitted deterministic controls.
- Change-impact recall ≥ 0.95 .
- Every public claim maps to a passing release-gate test.

6.12 M12 - VEIP Runtime and Evidence Adapter

Attribute	Design
Build decision	Provisional project-controlled reuse; verification dossier required
Foundation	VEIP and Authorization Evidence Pack project artifacts through a pinned adapter
Technical problem	A compiled control is valuable only if the runtime can bind an action to the exact current authority, control, facts, evidence, warrant, execution, reliance, and correction record.
Performance/capacity	Local runtime p95 ≤ 25 ms excluding external evidence retrieval; hosted control-plane lineage p95 ≤ 500 ms.

Responsibility

- Expose admission/evaluation API for proposed actions.
- Bind actor, action, resource, context, evidence, envelope, warrant, and policy state.
- Return structured operational verdict and reason codes.
- Create evidence pack and replay record.
- Record execution outcome, reliance, revocation, and correction links.
- Support envelope suspension and rollback.

Interface contract

Inputs

- Runtime request
- Published envelope

- Grounding facts
- Warrant artifact
- Evidence references
- VEIP policy

Outputs

- RuntimeEvaluation
- AuthorizationEvidencePack
- ReplayRecord
- RelianceRecord
- CorrectionRecord

Guarantees

- No terminal execution from proposal alone.
- Reliance requires recorded state.
- Correction preserves history and invalidates unsafe reliance.
- Replay binds exact versions and reproduces verdict where inputs are identical.
- Expiry/revocation propagate.

Explicit non-responsibilities

- Interpret documents.
- Originate authority.
- Represent DENY as proven prohibition unless grounds establish that fact.
- Execute business action itself in the reference build.

Failure modes

Failure	Required response
Envelope unavailable/stale	Fail closed or use explicitly approved cached version policy.
Evidence missing	DENY/CANNOT/ESCALATE per envelope.
Warrant mismatch	Reject evaluation.
Evidence store unavailable	Do not claim recorded state; execution policy determines fail closed.
Correction propagation delay	Suspend affected reliance and alert.

Security controls

- mTLS/service identity for runtime calls in pilot.
- Signed envelope and evidence-pack verification.
- Minimize evidence payload; reference protected records.
- Append-only audit and tamper detection.

Acceptance tests

- Dependency dossier passes the requirements in Section 3.5 before release freeze.
- POST /admit returns structured ALLOW/DENY/CANNOT/ESCALATE/NOT_APPLICABLE.
- Replay reproduces exact verdict and reason codes for pinned input.
- Revocation changes subsequent verdict and identifies affected reliance.
- Evidence pack traces to source and admission.

7. Canonical Data Model, Interfaces, and Versioning

7.1 Canonical entities

The reference implementation MUST define versioned JSON Schemas for:

- SourceDocument, SourceNode, SourceAnchor, SourceReference;
- AuthorityRecord, AuthorityRelationship, AuthorityIssue;
- CandidateNormativeUnit, InterpretationAlternative, ExtractionDiagnostic;
- UnresolvedQuestion, ReviewDocket, ReviewDisposition, AdmissionRecord;
- InstitutionalIRNode, InstitutionalIREdge, IRSnapshot;
- WarrantArtifact, MissingGround, ContradictionRecord;
- OpenControlEnvelope, EnvelopeManifest, HumanControlMatrix;
- TargetArtifact, TargetSourceMap, CompilationManifest;
- GeneratedTest, ConformanceReport, ReleaseGateResult;
- RuntimeEvaluation, AuthorizationEvidencePack, RelianceRecord, ChangeImpactRecord, CorrectionRecord.

Every enforceable artifact MUST contain stable ID, semantic version, source anchors, source hashes, authority basis, interpretation state, epistemic state, lifecycle state, admission reference, compiler version, test-suite hash, effective period, supersession/revocation references, and integrity digest.

7.2 Candidate normative unit example

```
{
  "$schema": "https://openinstitutional.org/schemas/candidate-normative-unit/v1.json",
  "unit_id": "cnu_PROC_0043",
  "source_anchor_ids": ["src_policy_2026_p14_c4_3"],
  "kind": "POWER",
  "actor": {"role_ref": "role_regional_director", "support": ["anchor_actor"]},
  "action": {"verb": "approve", "object_type": "contract", "support": ["anchor_action"]},
  "conditions": [
    {"expression_candidate": "contract.value <= delegated_limit", "support": ["anchor_threshold"]}
  ],
  "exceptions": [],
  "evidence_requirements": ["delegation_record"],
  "interpretation_state": "extracted",
  "epistemic_state": "uncertain",
  "alternatives": ["alt_PROC_0043_A", "alt_PROC_0043_B"],
  "model": {"provider": "declared-provider", "model_id": "declared-model", "prompt_version": "extract-v1"},
  "artifact_hash": "sha256:..."
}
```

7.3 Admission record example

```
{
  "$schema": "https://openinstitutional.org/schemas/admission-record/v1.json",
  "admission_id": "adm_01JZ...",
  "candidate_unit_id": "cnu_PROC_0043",
  "selected_interpretation_id": "alt_PROC_0043_B",
  "disposition": "ADMIT_WITH_QUALIFICATION",
  "qualification": "Applies only to regional purchases funded from the operating budget.",
  "permitted_uses": ["runtime_enforcement", "audit_testing"],
  "scope": {"organization_units": ["Region-East"], "jurisdiction": "enterprise-internal"},
  "reviewer": {"subject_id": "usr_...", "role": "Authority Steward", "authority_record_id": "auth_..."},
  "source_snapshot_hash": "sha256:...",
  "candidate_hash": "sha256:...",
  "admitted_at": "2026-07-28T18:00:00Z",
  "effective_from": "2026-08-01T00:00:00Z",
  "signature": "sigstore-or-enterprise-signature-reference"
}
```

7.4 Open Control Envelope example

```
{
  "$schema": "https://openinstitutional.org/schemas/open-control-envelope/v1.json",
  "control_id": "PROC-APPROVAL-017",
  "version": "1.3.0",
  "control_type": "DETERMINISTIC_ENFORCEMENT",
  "actor": {"role": "regional_director"},
  "action": {"verb": "approve", "resource_type": "contract"},
  "authority_chain": ["auth_board_policy", "auth_cfo_delegation", "auth_regional_director"],
  "conditions": [
    {"fact": "contract.value", "op": "<=", "value_from": "authority.delegated_limit"},
    {"fact": "conflict_check.status", "op": "==", "value": "clear"},
    {"fact": "required_documents.present", "op": "==", "value": true}
  ],
  "required_evidence": ["delegation_record", "conflict_check_record", "document_manifest"],
  "on_missing": "CANNOT",
  "on_unknown": "ESCALATE",
  "on_conflict": "DENY",
  "on_error": "DENY",
  "effective_period": {"from": "2026-08-01T00:00:00Z", "until": null},
  "source_anchor_ids": ["src_policy_2026_p14_c4_3", "src_matrix_Limits_E12"],
  "admission_ids": ["adm_01JZ..."],
  "ir_snapshot_hash": "sha256:...",
  "test_suite_hash": "sha256:...",
  "compiler": {"name": "oic", "version": "0.1.0"},
  "integrity": {"digest": "sha256:...", "signature": "..."}
}
```

7.5 Primary APIs

Method and path	Purpose	Safety condition
POST /v1/projects	Create isolated project	Tenant authorization required
POST /v1/projects/{id}/sources	Upload and register source	Malware/type/size checks before persistence
POST /v1/projects/{id}/compile	Start compilation to candidate stage	Cannot publish or admit
GET /v1/projects/{id}/docket	Retrieve prioritized review docket	Source permissions filter excerpts
POST /v1/review-items/{id}/dispositions	Create review disposition	Reviewer role and current source snapshot verified
POST /v1/projects/{id}/envelopes	Generate envelope from admitted snapshot	All required admissions and authority states valid
POST /v1/envelopes/{id}/targets/{target}	Compile target artifact	Conformance tests required before publish
POST /v1/envelopes/{id}/publish	Publish signed envelope/bundle	Publisher role separate; release gates pass
POST /v1/admit	Evaluate proposed runtime action	Exact published envelope and current evidence required
GET /v1/lineage/{artifact_id}	Return source-to-consequence graph	Tenant/project authorization
POST /v1/sources/{id}/versions	Register amendment/new version	Triggers impact analysis; does not overwrite
POST /v1/envelopes/{id}/revoke	Suspend or revoke envelope	Correction/revocation authority and reason required

7.6 Event contracts

- source.registered MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- source.parsed MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.

- authority.updated MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- extraction.completed MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- docket.generated MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- review.disposition_recorded MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- ir.snapshot_admitted MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- envelope.generated MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- target.compiled MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- envelope.published MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- runtime.evaluated MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- source.version_changed MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- impact.detected MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- envelope.suspended MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- reliance.flagged MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.
- correction.recorded MUST include event ID, tenant/project, aggregate ID/version, occurred_at, producer version, trace ID, artifact digests, and schema version.

7.7 Error taxonomy

Prefix	Class	Examples	Default operational behavior
SRC	Source/parse	SRC_UNSUPPORTED_FORMAT, SRC_OCR_LOW_CONFIDENCE	Stop affected source; preserve project
AUTH	Authority	AUTH_MISSING_BASIS, AUTH_EXPIRED, AUTH_CONFLICT	Block dependent admission/publication
SEM	Semantic extraction	SEM_UNSUPPORTED_FIELD, SEM_SOURCE_MISMATCH	Keep candidate non-executable
REV	Review/admission	REV_STALE_SOURCE, REV_UNAUTHORIZED	Reject disposition
IR	Canonical model	IR_DANGLING_EDGE, IR_INVALID_STATE	Reject snapshot/publication
ZTL	Warrant	ZTL_CANNOT, ZTL_CONTRADICTION, ZTL_VERSION_MISMATCH	CANNOT/block warrant-dependent publication
ENV	Envelope	ENV_UNOBSERVABLE_INPUT, ENV_STALE_DEPENDENCY	Non-deployable/suspend
TGT	Target	TGT_COMPILE_ERROR, TGT_BEHAVIOR_MISMATCH	Block target publication
RUN	Runtime	RUN_EVIDENCE_MISSING, RUN_ENVELOPE_REVOKED	DENY/CANNOT/ESCALATE per contract
SEC	Security	SEC_MALWARE, SEC_TENANT_VIOLATION	Quarantine/deny/alert
SYS	Platform	SYS_TIMEOUT, SYS_DEPENDENCY_UNAVAILABLE	Bounded retry; fail visible

7.8 Versioning

- Schema versions use semantic versioning. A major version changes interpretation or compatibility; minor adds backward-compatible optional capability; patch fixes defects without semantic change.
- Source versions use immutable content hashes plus institution-defined version/effective metadata.
- IR snapshots are immutable. New admission or source change creates a new snapshot.
- Envelope versions increment major for incompatible decision semantics, minor for backward-compatible scope/feature additions, and patch for non-semantic metadata/tooling fixes.
- Target artifacts bind exact envelope and adapter versions; recompilation under a new adapter creates a new artifact even if behavior is equivalent.
- Benchmark releases pin corpus, annotations, baseline commits, metric implementation, thresholds, and hardware/configuration profile.

8. Technology Stack and Deployment Architecture

8.1 Reference stack

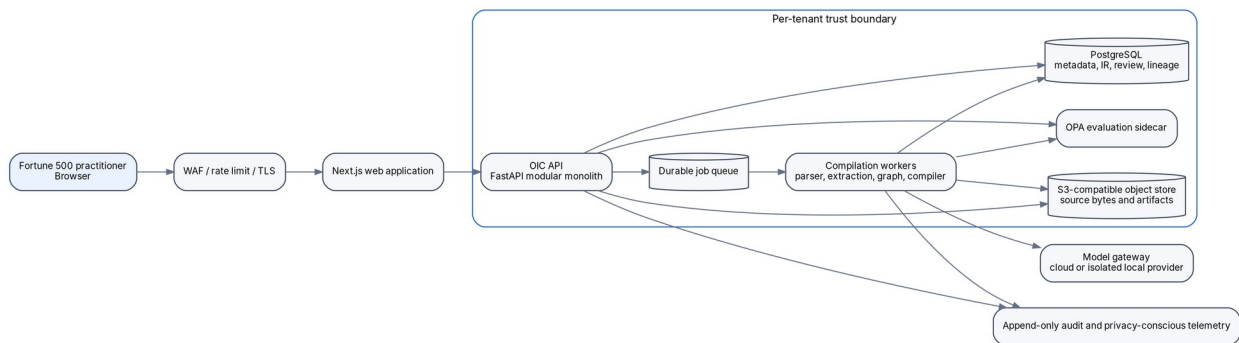
Layer	Selected technology	Rationale / constraint
Language	Python 3.12	Strong document/AI ecosystem, rapid reference implementation, typed validation; pin patch version at code freeze.
API	FastAPI + Pydantic v2	OpenAPI, typed contracts, async jobs, JSON Schema alignment.
Web UI	React + TypeScript; Next.js stable release pinned at code freeze	Accessible practitioner workflow and artifact inspection.
Relational store	PostgreSQL 17 with JSONB and adjacency tables	Transactional review/admission, mature operations, enough graph capability for reference/pilot.
Object store	S3-compatible storage; MinIO for local/on-prem	Immutable source bytes, derivative artifacts, signed manifests.
Job orchestration	Durable PostgreSQL-backed queue for reference; pluggable enterprise queue later	Avoid unnecessary infrastructure while preserving retry/idempotency contracts.
Document conversion	Docling adapter	Format/layout capability behind sandboxed OIC interface.
Model gateway	Provider-agnostic structured-generation adapter	No semantic dependency on one model; local model profile required for offline path.
Logical warrant	ZTL pinned adapter	Deterministic T/F/Z and dependency artifacts.
Policy target	OPA/Rego pinned release	Primary executable target and test runtime.
Business representation	DMN 1.5 export profile	Readable decision tables for suitable deterministic subsets.
Observability	OpenTelemetry-compatible traces, metrics, logs	Vendor-neutral operations; no source text in default telemetry.
Packaging	Docker Compose for local; Helm/Kubernetes profile for pilot	One-command experience and enterprise deployment path.
Supply chain	SBOM, SLSA provenance, Sigstore/Cosign-compatible signing	Release integrity and dependency accountability.

8.2 Why not a distributed microservice architecture in week one

A twelve-service implementation would consume the week on networking, deployment, tracing, retries, authentication, and schema coordination without improving the semantic proof. The modular monolith isolates responsibilities through packages, immutable artifacts, typed interfaces, job boundaries, and role checks. Extraction into services is triggered only by measured needs such as independent scaling, security isolation, deployment boundaries, or separate ownership.

The parser and untrusted model execution already run in isolated workers because their security and resource profiles justify process boundaries. The policy runtime may be embedded or separate depending on deployment.

8.3 Hosted deployment

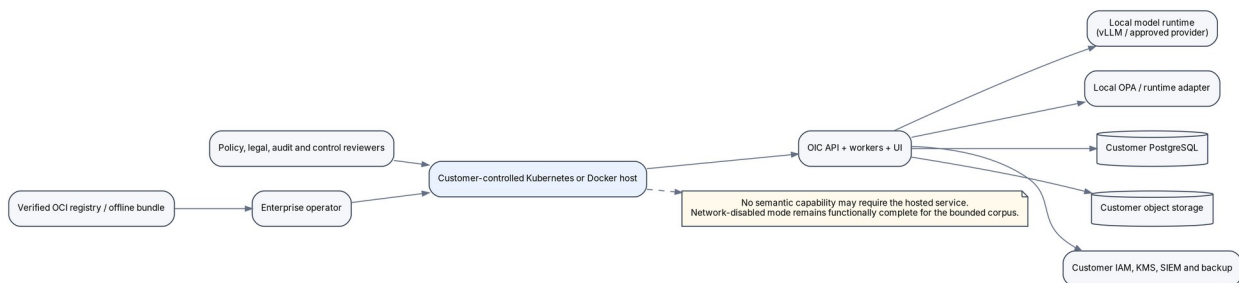


Hosted Institutional Compiler Lab deployment.

Hosted Lab requirements:

- tenant-scoped projects and encryption context;
- public/synthetic/non-confidential documents only at launch;
- file validation and malware scanning before parsing;
- isolated parser/model workers;
- explicit model/provider disclosure per run;
- no model training on uploaded content;
- downloadable source graphs, candidate units, docket, admissions, envelope, target artifacts, tests, lineage, limitations, and benchmark report;
- default deletion within 24 hours for unauthenticated/sample projects; configurable shorter retention;
- rate and cost limits; no automatic deployment to customer production runtimes;
- same compiler commit and schemas as the open release.

8.4 On-premises and air-gapped deployment



On-premises and offline deployment profile.

The enterprise path MUST support a customer-controlled environment. The compiler, review console, source storage, database, ZTL engine, target compiler, OPA runtime, and evidence store operate without a hosted semantic dependency. Model options include an approved local model or a customer-controlled private endpoint. Release packages include signed images, SBOMs, model/configuration manifests, offline documentation, and fixture-based acceptance tests.

Air-gapped deployment is not considered complete merely because containers start. The TDD requires proof that no module attempts outbound network access during compilation, review, publication, runtime evaluation, or replay under the offline profile.

8.5 Data storage model

Store	Data	Retention baseline	Integrity
Object store	Original files, parser artifacts, rendered views, target bundles, benchmark artifacts	Project policy; public Lab short-lived	Content hash, immutable/versioned object, encryption
PostgreSQL	Metadata, graph edges, reviews, admissions, jobs, role bindings, indexes	Project policy	Transactions, row-level tenant controls, append-only admission/event records

Store	Data	Retention baseline	Integrity
Evidence store	Runtime evidence references, AEP, replay, reliance, correction	Customer regulatory policy	Signed manifests, WORM/append-only option
Telemetry backend	Metrics, traces, redacted structured logs	30-90 days planning baseline	No source content; access and retention controls
Secret/KMS store	Keys, provider credentials, signing identities	Rotation policy	Hardware/cloud KMS or customer equivalent

9. Institutional Compiler Lab: Practitioner Experience

The Lab is a transparent evaluation environment, not a magical compliance chatbot. The primary navigation is deliberately limited to five stages:

1. **Upload** - establish corpus and source status.
2. **Inspect** - view structural recovery, authority records, candidate meaning, and limitations.
3. **Resolve** - answer the prioritized docket and record institutional disposition.
4. **Compile** - generate and download envelope, control matrix, Rego, tests, and manifests.
5. **Test** - change actor, action, amount, date, evidence, or authority state and observe verdict and lineage.

9.1 Entry points

- **Inspect a complete sample:** No upload required. Uses a complex procurement corpus with known conflicts, amendment, expired delegation, and human-judgment boundary.
- **Upload sample documents:** Public Lab accepts only content the user is authorized to disclose and that is non-confidential, non-regulated, and non-privileged.
- **Run locally:** Direct link to repository and one-command demo.

9.2 Project summary

The platform reports transformation counts without implying correctness:

72 source clauses → 41 candidate normative units → 33 candidate controls → 11 evidence duties → 6 blocking questions → 4 human-judgment boundaries → 2 unresolved authority conflicts.

Each number is clickable. “Candidate” and “admitted” are visually distinct. A high extraction confidence is never labeled approved.

9.3 Review card contract

Every review card displays:

- exact source excerpt and location;
- document status and authority context;
- proposed plain-language operational meaning;
- actor, action, conditions, exceptions, evidence, time, and escalation;
- alternative interpretations;
- one allowed, one denied, and one unknown example where applicable;
- affected controls and consequence level;
- known parser/model limitations;
- actions: admit, qualify, modify, reject, escalate, non-automatable.

The default view is understandable to a policy practitioner. Formal graph and JSON views are available but secondary.

9.4 Runtime playground

The user can vary:

- actor role and delegation state;
- transaction/action type;
- monetary or quantitative threshold;
- evidence present/missing/contested;
- date and policy version;
- emergency exception state;
- revocation and amendment state.

The response shows operational verdict, reason codes, missing facts, exact control/envelope version, ZTL warrant summary, evidence-pack reference, and full lineage.

9.5 Honest limitations in the interface

Limitations are generated at project, clause, control, target, and runtime levels. Examples:

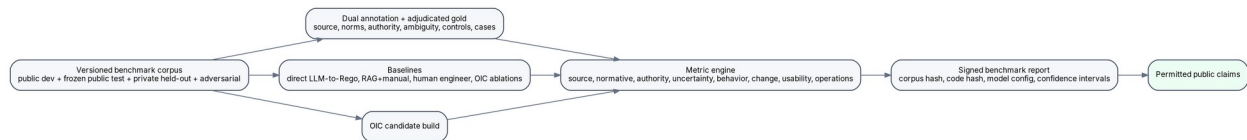
- “The phrase reasonable efforts has no admitted evaluation procedure and cannot be compiled into a deterministic outcome.”
- “Delegation scope is not established by the supplied corpus. This control remains non-deployable.”
- “The scanned approval matrix contains two low-confidence cells requiring correction.”

- "DMN export omits this evidence-retention obligation because it is not a decision rule; the obligation remains in the envelope."

9.6 Accessibility and usability requirements

- WCAG 2.2 AA target for hosted Lab and reference UI.
- Complete keyboard operation for upload, inspect, review, compile, test, drawers, and state transitions.
- No status communicated by color alone.
- Plain-language labels with expandable technical definitions.
- Source excerpts remain selectable and downloadable; no image-only critical content.
- Tables have headers, captions, and responsive alternatives.
- Timed usability study with Fortune 500 practitioner profiles; results published with limitations.

10. OIC-Bench: Evaluation Corpus, Metrics, and Claims



Benchmark pipeline from frozen corpora to permitted claims.

OIC-Bench is a governed artifact equal in importance to the compiler. It evaluates the category OIC proposes, not merely whether generated Rego compiles. Version 1.1 separates the **week-one benchmark preflight** from the **full v0.1 benchmark**. No result from the preflight may be presented as a full benchmark or market-superiority result.

10.1 Benchmark stages and corpus partitions

10.1.1 Seven-day benchmark preflight subset

The experimental reference slice uses a deliberately reduced, pinned subset to validate metric implementations and expose obvious architecture failures. Proposed minimum composition:

- 4 public or synthetic documents;
- at least 25 pages;
- at least 35 adjudicated normative clauses;
- at least 12 candidate controls;
- at least 6 known ambiguities, conflicts, or missing-authority cases;
- at least 20 runtime transaction cases;
- at least 3 source-version or amendment pairs;
- at least one table, annex, exception, expired delegation, discretionary clause, and malicious document instruction.

These figures are **founder-selected engineering scope estimates**, not measured sufficiency thresholds. The preflight is complete when the metric code, manifests, baseline harness, and artifact lineage run reproducibly. It does not establish statistical reliability, enterprise generalization, or superiority.

10.1.2 Full OIC-Bench v0.1

Partition	Initial domain	Proposed minimum composition	Use
Public development	Procurement controls	7 documents; 52 pages; visible annotations	Engineering, examples, PR tests; not valid for final quality claims
Public frozen test	Financial/approval controls	8 documents; 48 pages; annotations released under freeze policy	Reproducible public evaluation
Private held-out	Audit/assurance controls	6 documents; 40 pages; controlled gold labels	Independent release evaluation and contamination resistance
Adversarial	Mutated policy fragments and files	60 semantic/security mutations	Prompt injection, negation, thresholds, time, actor, exception, authority, and parser attacks
Usability	Review docket/control tasks	12-20 tasks across practitioner roles	Orientation, review time, understanding, agreement, and error rate

Proposed minimum aggregate content:

- 21 source documents and 140 pages;
- at least 260 gold normative clauses;
- at least 85 candidate controls;
- at least 34 known ambiguities/conflicts;
- at least 12 temporal, amendment, or supersession relations;
- at least 160 runtime transaction cases;
- at least 18 source-version pairs for change-impact evaluation;
- tables, annexes, definitions, cross-references, exceptions, evidence duties, discretionary language, and expired authority.

Every corpus release **MUST** include provenance, license/reuse rights, file hashes, annotation schema, annotator role, adjudication procedure, known bias, contamination analysis, and a signed manifest. Corpus scale and composition are pre-registered targets subject to calibration before freeze.

10.2 Held-out and contamination policy

- Development data and gold labels are visible to engineers.

- Frozen public test data may be visible, but implementation changes after inspecting failures must be reported and a new held-out run required for release claims.
- Private held-out annotations are access-controlled and not included in model prompts.
- Baseline and OIC runs use the same source inputs and declared model where comparison requires isolating architecture.
- Possible pretraining exposure is documented; the benchmark emphasizes transformations, conflicts, amendments, and institution-specific combinations that cannot be assumed uncontaminated solely from public text.
- A future corpus refresh is triggered by material domain drift, benchmark saturation, new document types, or annual review. Historical versions remain available.
- Annotator selection, inter-annotator agreement, adjudication time, and conflicts of interest MUST be disclosed.

10.3 Metric families

Metric	Definition	Why it matters
Source-supported executable fields	Executable envelope fields with defensible source/admission support / all executable fields	Prevents invented controls
Unsupported executable-field rate	Executable fields without sufficient source/admission support / all executable fields	Primary harm metric
Normative Field macro-F1	Macro-F1 across actor, action, modality, condition, exception, evidence, time, escalation, and discretion	Measures operational-meaning decomposition
Authority Reconstruction F1	F1 over authority source, issuer, scope, delegation, precedence, and period	Differentiating authority metric
Expired/superseded recall	Correctly detected inactive authority / all gold inactive authority	Prevents stale authorization
Ambiguity recall	Gold blocking ambiguities detected / all gold blocking ambiguities	Rewards visible uncertainty
False-resolution rate	Ambiguous/unknown items incorrectly converted into determinate controls / all ambiguous/unknown items	Penalizes confident guessing
Unknown-to-false conversion	Count of Z/unknown states serialized as false	Must equal zero
Behavioral conformance	Correct runtime outcomes / gold deterministic cases	Tests enforceable behavior
Change-impact recall	Affected controls correctly identified / all gold affected controls	Measures continuous compilation
Cross-target equivalence	Cases with equivalent outcome/reason class across supported targets / all shared cases	Protects target independence
Reviewer time to admission	Median active time per review item after orientation	Measures usability bottleneck
Time to first envelope	Elapsed time from accepted corpus to first reviewable/deployable envelope	Customer outcome

10.4 Proposed full-benchmark gates

The following values are **pre-registered proposed safety and quality gates selected before implementation**. They are not observed results and do not imply that the corpus is sufficient. Before full freeze, the benchmark lead must document calibration evidence and the design authority must either confirm or amend each value through an ADR without reference to hidden held-out outcomes.

Gate	Proposed threshold	Evidence classification / note
Source-supported executable fields	$\geq 99\%$	Proposed safety gate; frozen test and held-out combined
Unsupported executable-field rate	$\leq 1\%$	Proposed safety gate; any unsupported high-consequence authority field is critical regardless of average
Normative Field macro-F1	≥ 0.85	Proposed quality gate; publish per-field values
Authority Reconstruction F1	≥ 0.80	Proposed bounded-corpus quality gate
Expired/superseded recall	≥ 0.95	Proposed critical safety gate
Ambiguity recall	≥ 0.85	Proposed uncertainty-integrity gate
False-resolution rate	≤ 0.05	Proposed safety gate; lower is better
Unknown-to-false conversion	0	Architectural invariant, not an empirical average

Gate	Proposed threshold	Evidence classification / note
Behavioral conformance	≥ 0.95	Proposed gate for admitted deterministic controls
Change-impact recall	≥ 0.95	Proposed gate on version-pair corpus
Cross-target equivalence	≥ 0.98	Deferred until a second executable target ships
Reviewer orientation	≤ 15 minutes	Proposed usability target; requires real participants
Median review time	≤ 3 minutes/item	Proposed usability target; report distribution and task complexity
Time to first envelope	≤ 4 hours	Proposed reference-corpus target excluding reviewer scheduling delay
Local runtime latency	p95 ≤ 25 ms	Engineering target for OPA evaluation excluding external evidence retrieval

10.5 Baselines

Baseline	Implementation requirement	Fair comparison boundary
Direct LLM-to-Rego	Same model, source corpus, and transaction tests; prompt generates Rego directly	Compare compile, source support, ambiguity, behavior, and change impact; disclose prompt
Modular document-to-policy pipeline	Detection, extraction, validation, target compilation, and tests without OIC admission/authority IR	Compare shared capabilities; report OIC-only dimensions separately
Human policy engineer	Experienced practitioner creates control matrix and target policy using ordinary tools	Compare elapsed/active time, defects, source coverage, ambiguity found, and change update
Source-grounded knowledge graph	Extract source entities/relations without formal admission/envelope	Test whether graph alone solves enforceable meaning
RAG plus rule generation	Retrieve source passages and generate target policy per request	Test target-specific reinterpretation and change consistency
Commercial capability review	Public documentation and, where available, controlled product trial under a disclosed scenario	Do not equate missing public documentation with missing product capability; report unverifiable dimensions as unknown

10.6 Comparative-claim protocol

No comparative outperformance claim is permitted from the seven-day preflight. A claim may be considered only after the full benchmark, baseline implementations, annotation protocol, and held-out run are frozen and independently reviewed.

The following are **provisional founder-selected claim thresholds for later calibration**, not current product claims:

- at least 50% relative reduction in unsupported executable-field rate versus direct LLM-to-Rego;
- at least 20 percentage-point improvement in blocking ambiguity recall;
- at least 30 percentage-point improvement in change-impact recall;
- at least 40% reduction in active human time to first admitted envelope versus the manual baseline, with no more than 2 percentage-point loss in adjudicated behavioral quality;
- all applicable absolute release gates pass.

Before use, each threshold **MUST** be classified as calibrated, retained as founder risk judgment, or withdrawn.

Every public comparison **MUST** name the baseline, version, model, corpus, metric implementation, uncertainty, and limitations. “Best,” “first,” “world-leading,” and universal-superiority language remains forbidden unless independently supportable under Appendix C.

10.7 Corpus rot and benchmark governance

- Freeze each corpus version with SHA-256 manifest and release tag.
- Retain old versions so historical claims remain reproducible.
- Review annually or when product distribution materially changes.
- Add new domains without removing difficult old cases.
- Publish annotation corrections as versioned errata and rerun affected claims.
- Separate compiler maintainers from held-out adjudicators where feasible.
- Require an external reproduction before claiming infrastructural maturity.
- Treat benchmark construction time, annotation disagreements, and failed gates as release evidence rather than schedule exceptions.

11. Security, Privacy, Compliance, and Abuse Resistance

Security and privacy are architectural constraints because policy corpora reveal institutional authority, control weaknesses, thresholds, incident procedures, and sensitive operations. Public and enterprise deployment profiles are intentionally different.

11.1 Data classification

Class	Examples	Default handling
Public	Published statutes, public manuals, synthetic sample corpus	Allowed in public Lab subject to rights
Internal	Ordinary enterprise policies, procedures, control matrices	Hosted enterprise or on-prem only; encrypted and access-controlled
Confidential	Delegations, security procedures, nonpublic findings, contract terms	On-prem/private deployment; restricted model/telemetry path
Regulated / sensitive personal	PII, financial records, health data, employee investigations	Not accepted by public Lab; customer-specific compliance controls
Privileged / export-controlled / classified	Legal privilege, national-security or export-controlled material	Outside public and default hosted scope; specialized deployment/authority required
Derived control intelligence	Authority graph, envelopes, exceptions, thresholds, findings	Classify at least as high as most sensitive contributing source

11.2 Threat model

ID	Threat	Primary controls	Inherent severity
T-01	Malicious PDF/document parser exploit	Sandbox, patched parser, seccomp/container limits, malware scan, no network	High
T-02	Prompt injection embedded in documents	Content-only model role, no tools/network, structured schema, provenance checks, adversarial tests	High
T-03	Cross-tenant data leakage	Tenant-scoped keys, row/object policies, authorization tests, separate namespaces	Critical
T-04	Unauthorized reviewer admission	RBAC/ABAC, SSO/MFA, authority record, signed append-only disposition	Critical
T-05	Administrator impersonates policy authority	Role separation, dual control for publisher/admin overrides, explicit audit	High
T-06	Forged/substituted source	Hash at ingestion, immutable original, signed manifest, version lineage	Critical
T-07	Manipulated authority metadata	Authority Steward role, evidence references, change approval, graph diff	Critical
T-08	Model/provider exfiltration	Approved endpoints, no-training terms, local option, data minimization, DLP controls	High
T-09	Hallucinated executable field	Source-support validator, admission gate, unsupported-field metric, fail closed	Critical
T-10	Stale envelope execution	Version checks, effective periods, revocation feed, runtime cache policy	Critical
T-11	Target compiler supply-chain compromise	Pinned toolchain, SBOM, signatures, sandbox, behavior tests	High
T-12	Policy rollback attack	Monotonic publication records, signed versions, rollback authorization and visibility	High
T-13	Signing-key compromise	KMS/HSM, rotation, revocation, transparency/audit, limited signing identity	Critical
T-14	Benchmark manipulation	Pinned corpus/metrics, separation of duties, signed reports, external reproduction	Medium
T-15	Poisoned community policy pack	Sandbox, provenance, review, maturity levels, no auto-trust	High

ID	Threat	Primary controls	Inherent severity
T-16	Sensitive source text in logs/telemetry	Structured redaction, allowlist fields, no prompt/source bodies, audits	High
T-17	Denial of service / decompression bomb	File/page/token/CPU limits, queues, quotas, circuit breakers	High
T-18	Runtime fact spoofing	Service identity, evidence signatures, source systems of record, freshness checks	Critical
T-19	Evidence deletion/tampering	Append-only/WORM option, digests, replication, retention policy	Critical
T-20	Correction fails to reach reliance	Dependency graph, suspension, notification, reconciliation job, incident gate	Critical

11.3 Prompt-injection posture

Policy documents are data, never instructions to the system. The model stage receives a fixed extraction task, no tool credentials, no browser, and no ability to alter source or admission records. Output is constrained to schema and validated against source anchors. Prompt-injection detection adds visibility but is not the trust boundary. The real controls are capability isolation, source-support checks, institutional admission, and target conformance.

11.4 Privacy and retention

- Public Lab launch accepts only public, synthetic, or non-confidential samples; the user must affirm disclosure authority.
- Unauthenticated/sample projects default to deletion within 24 hours; shorter immediate deletion is available.
- Enterprise retention is customer-configurable and must include legal hold/deletion semantics where required.
- Customer content is not used for training or product-wide evaluation without a separate explicit agreement.
- Telemetry excludes source text, prompts, evidence payloads, and personal identifiers by default.
- Download and deletion operations are recorded; derived artifacts are deleted consistently with source policy unless retention is independently required.

11.5 Secure development and supply chain

- NIST Secure Software Development Framework practices guide planning, development, verification, and response.
- OWASP prompt-injection guidance informs the AI threat model, but model isolation remains primary.
- Every release includes dependency lockfiles, SBOM, vulnerability scan, signed containers/artifacts, and provenance.
- SLSA-compatible provenance and Sigstore/Cosign-compatible signing are target controls for public releases.
- Secrets never enter repository, benchmark fixtures, generated artifacts, or logs.
- Security disclosure policy and response SLA are published in SECURITY.md.

12. Reliability, Failure Isolation, Rollback, and Operations

12.1 Reliability classes

- **Authoring/compilation path:** asynchronous and resumable; convenience may degrade, but admitted artifacts must never be silently corrupted.
- **Publication path:** consistency and integrity dominate latency; fail closed.
- **Runtime evaluation path:** low latency and high availability; uses only published signed envelopes and declared cache/fallback policy.
- **Evidence/continuity path:** durability and replay dominate; absence of required record prevents a claim that execution is recorded or safe to rely upon.

12.2 Service objectives

Profile	Availability target	RTO	RPO	Notes
Public Lab reference	99.5% monthly after public release	4 hours	1 hour	No production enforcement dependency
Enterprise beta control plane	99.9% monthly	1 hour	15 minutes	Subject to customer deployment profile
Local runtime reference	99.95%	Customer topology	Published bundle/config	Evaluation excludes external evidence latency
Evidence store pilot	99.9% with durability target defined by backend	1 hour	<=15 minutes or WORM replication policy	No claim beyond selected storage guarantees

12.3 Idempotency and retries

- Upload registration keyed by project, file digest, and idempotency key.
- Compilation stages keyed by immutable input artifact hashes and stage configuration.
- Review dispositions are non-idempotent semantic events but accept client idempotency keys to prevent duplicate submission.
- Envelope generation and target compilation are deterministic where feasible and create new artifacts only when inputs/config change.
- Runtime evaluation IDs prevent duplicate evidence packs for retried calls; business action execution remains outside reference adapter.
- Retries use exponential backoff, jitter, bounded attempts, and dead-letter visibility. No infinite retry storms.

12.4 Kill switches and rollback

Scope	Kill switch / rollback behavior
Model provider	Disable provider/profile; preserve prior artifacts; route to approved alternative or pause extraction
Compiler version	Block new use; revert to last approved version; affected artifacts flagged
Target adapter	Suspend publication for adapter; retain prior signed bundle if still valid
Project/tenant	Read-only or fully suspended mode; no publication/runtime changes
Envelope/control	Suspend or revoke exact version; runtime denies/cannot according to policy
Hosted Lab	Disable uploads/compilation while preserving read/download where safe
Review system	Review-only export or pause admission; no admin bypass
Evidence path	Block executions requiring recorded state or operate under explicit preapproved degraded policy

12.5 Blast-radius boundaries

Failures must be containable by tenant, project, corpus, source, IR snapshot, envelope, control, compiler version, model profile, and target adapter. A parser or extractor defect cannot alter already admitted artifacts. An adapter

defect cannot alter canonical IR. A customer corpus cannot influence another tenant. A revoked envelope cannot be restored by rolling back the application database without detecting integrity/version mismatch.

12.6 Change propagation



Source change, impact analysis, suspension, re-review, publication, and correction.

A source version change creates a new immutable node and triggers graph traversal. The system identifies affected definitions, interpretations, authority relations, warrants, envelopes, target artifacts, tests, runtime caches, and reliance records. Controls with material unresolved impact become suspended or marked re-review-required before a new version is published. Historical versions remain replayable.

12.7 Observability and on-call

- OpenTelemetry trace from upload/job through parsing, extraction, docket, review, compilation, publication, runtime, and lineage.
- Metrics for stage latency, queue age, failure rate, parser/model cost, source-support failures, review backlog, stale envelopes, runtime verdicts, and correction propagation.
- Logs contain IDs, versions, reason codes, and hashes - not source or evidence bodies.
- Alerts prioritize unauthorized publication, cross-tenant access, stale/revoked runtime use, signing failure, evidence durability, and correction backlog.
- Runbooks exist for parser compromise, model regression, benchmark invalidation, target mismatch, key compromise, and stale-envelope incident.

13. Capacity, Performance, Cost, and Resourcing

All numerical statements in this section **MUST** be classified as **measured result**, **externally sourced reference**, **engineering estimate**, **pre-registered target**, **release gate**, or **illustrative scenario**. Unclassified numbers may not appear in release claims.

13.1 Evidence classification for planning numbers

Number type	Meaning	Permitted use
Measured result	Produced by a pinned run or observed delivery record	May support a scoped claim with method and uncertainty
Externally sourced reference	Derived from a cited public or contractual source	May support planning with source limitations
Engineering estimate	Team judgment before measurement	Planning only; must display confidence and calibration date
Pre-registered target	Value selected before implementation or held-out evaluation	Defines success/failure; is not evidence of current performance
Release gate	Approved threshold that blocks a named release state	Controls release; does not imply broader generalization
Illustrative scenario	Example used to expose cost or capacity relationships	Must not be represented as forecast or quote

13.2 Capacity profiles

Profile	Proposed corpus limit	Proposed concurrency	Primary purpose	Classification
Public Lab alpha	<=100 pages/project; <=50 MB/file; <=10 active compile jobs	<=100 concurrent readers/reviewers planning ceiling	Demonstration and evaluation	Engineering estimate
Local reference	<=500 pages/project on documented workstation	1-4 compile jobs	Open reproduction and bounded pilot	Engineering estimate
Enterprise pilot candidate	<=5,000 pages/corpus; configurable file limits	Tenant-sized workers and queues	Controlled institutional corpus	Engineering estimate pending load tests
Runtime evaluator	Published envelopes up to 10,000 controls per bundle	Scale horizontally with OPA deployment	Low-latency decision path	Illustrative planning ceiling

13.3 Performance budgets

Operation	Proposed reference budget	Classification
Upload registration	p95 <=1 second excluding client transfer	Engineering target
50-page digital parse	p95 <=5 minutes	Engineering target
50-page scanned parse	p95 <=15 minutes	Engineering target
50-page candidate extraction	p95 <=15 minutes	Engineering target
Docket generation for 1,000 units	<=10 seconds	Engineering target
Review card load	p95 <=2 seconds	Engineering target
Lineage expansion	p95 <=500 ms	Engineering target
Envelope generation for 1,000 controls	<=15 seconds	Engineering target
Target compilation for 1,000 controls	<=20 seconds	Engineering target
Local OPA evaluation	p95 <=25 ms excluding external evidence	Proposed release gate after measurement
Full public benchmark	<=60 minutes on documented hardware	Engineering estimate; not a week-one commitment
PR preflight suite	<=15 minutes	Engineering target

13.4 Model and infrastructure cost

A 50-page digital policy corpus may produce approximately 75,000-200,000 input tokens and 20,000-60,000 output tokens across extraction, alternatives, question generation, and examples, depending on layout and chunking. The corresponding **\$5-\$50 per 50-page corpus** range is an **illustrative engineering estimate**, not a measured result, quote, or benchmark claim. It **MUST** be replaced or retained with calibration evidence after the first ten reproducible runs.

The system **SHOULD** use lower-cost or local models for classification and candidate generation where benchmark quality permits, reserving stronger models for disputed or high-impact clauses. Cost optimization cannot reduce source support, ambiguity preservation, admission requirements, or reproducibility.

13.5 Seven-day experimental team

Role	Proposed FTE for seven-day slice	Primary responsibility
Technical lead / design authority	1.0	Architecture, contracts, integration decisions, scope and claims control
Compiler/IR engineers	2.0	Schemas, IR, envelope, target compilation
Document/LLM engineer	1.0	Ingestion, extraction, source support
ZTL/VEIP adapter engineer	1.0	Provisional warrant and continuity adapters; dossiers
Full-stack/product engineer	1.0	Lab workflow and artifact inspection
Benchmark/QA engineer	1.0	Preflight corpus, baselines, metric code, tests
Security/DevOps engineer	1.0	Sandboxing, deployment, signing, security preflight

The **8 FTE / 56 person-day** figure is a staffing arithmetic assumption, not evidence that all listed roles are available or that the scope will complete. The **\$1,200-\$2,000 per person-day** loaded-rate range and resulting **\$67,000-\$112,000** internal-equivalent cost are founder planning estimates pending finance and market calibration. They exclude prior ZTL/VEIP work, legal review, paid datasets, external annotation, accessibility audit, penetration testing, and independent review.

13.6 Subsequent planning ranges

Horizon	Planning team/effort	Illustrative internal-equivalent range	Main additions	Classification
30-day reproducible alpha	8-12 FTE; approximately 160-240 person-days	\$170,000-\$360,000	Dependency dossiers, security hardening, open reproduction, hosted alpha, benchmark tooling	Founder planning estimate
60-day evidence release	Incremental 8-14 FTE plus domain reviewers	To be recalibrated after alpha	Full benchmark, annotation, baselines, usability, external technical/security review	Estimate intentionally not frozen
90-day enterprise pilot candidate	12-20 FTE; approximately 720-1,200 person-days cumulative scenario	\$576,000-\$1,200,000	Tenancy, SSO, on-prem, reliability, support, multi-corpus operations, agent integration	Illustrative planning scenario

13.7 Ongoing operational burden

- Parser, model, and target dependency updates with regression runs.
- Benchmark corpus governance, annotation corrections, and independent reproduction.
- Security vulnerability response, signing keys, SBOM/provenance, and incident handling.
- Hosted infrastructure, on-call, backups, evidence durability, and customer support.
- Schema/adapter compatibility and community contribution review.
- Domain-specific policy packs and practitioner onboarding.
- Claims review so public positioning does not outrun evidence.
- Ongoing verification of ZTL and VEIP project-controlled components or their replacements.

14. Open Repository, Governance, and Contribution Model

14.1 Repository contract

```

/
├── README.md
├── GOVERNANCE.md
├── CONTRIBUTING.md
├── SECURITY.md
├── THREAT_MODEL.md
├── ROADMAP.md
├── LIMITATIONS.md
├── CLAIMS.md
├── VERSIONING.md
├── LICENSE
├── docs/
├── schemas/
│   ├── source-document/
│   ├── authority/
│   ├── candidate-normative-unit/
│   ├── institutional-ir/
│   ├── admission-record/
│   ├── open-control-envelope/
│   └── evidence-pack/
├── oic/
│   ├── intake/
│   ├── source_model/
│   ├── authority/
│   ├── extraction/
│   ├── ir/
│   ├── docket/
│   ├── review/
│   ├── envelope/
│   ├── targets/
│   ├── tests/
│   └── continuity/
├── adapters/
│   ├── docling/
│   ├── ztl/
│   ├── opa/
│   ├── dmn/
│   └── veip/
├── platform/
├── ui/
├── benchmarks/
│   ├── corpora/
│   ├── annotations/
│   ├── baselines/
│   ├── metrics/
│   └── reports/
├── examples/procurement-complex/
├── tests/
├── adr/
└── .github/

```

14.2 Ten-minute contributor experience

```
git clone <repository-url>
cd open-institutional-compiler
docker compose up -d
oic compile examples/procurement-complex/
oic review --project examples/procurement-complex/project.json
oic envelope build --project examples/procurement-complex/project.json
oic target compile --target opa --project examples/procurement-complex/project.json
oic evaluate examples/procurement-complex/transactions/contract-017.json
oic benchmark run --suite public-pr
```

Expected outputs include source-graph.json, authority-graph.json, candidate-units.json, review-docket.json, admissions.json, institutional-ir.json, control-envelope.json, policy.rego, tests/, lineage.json, limitations.json, and a signed manifest.

14.3 Likely contributor communities

- OPA/Rego and Cedar engineers adding target adapters and conformance tests.
- Rules-as-Code, legal engineering, Catala, OpenFisca, and LegalRuleML practitioners improving normative models and examples.
- Document-processing contributors improving formats, layout, OCR, tables, and multilingual support.
- Compliance, audit, procurement, finance, and security practitioners contributing public/synthetic policy packs and gold review cases.
- Formal-methods and ZTL contributors improving warrant, contradiction, and bounded-proof artifacts.
- VEIP and runtime-governance contributors strengthening evidence, replay, reliance, revocation, and correction.
- Security engineers improving sandboxing, signatures, supply-chain integrity, and threat fixtures.
- Accessibility and localization contributors reducing the practitioner learning curve.

14.4 Open governance

The recommended baseline is Apache-2.0 for code and schemas, and CC BY 4.0 for original documentation and benchmark annotations where source licenses permit. Final licensing requires counsel and compatibility review, especially for third-party corpora and reused code. OIC must not copy incompatible code merely because an architecture is useful.

Governance artifacts define maintainer roles, decision process, code of conduct, security disclosure, semantic versioning, schema evolution, benchmark submissions, adapter maturity, conformance marks, release signing, trademark use, and conflict-of-interest disclosure.

Semantic changes to canonical schemas, state models, admission rules, authority behavior, benchmark thresholds, or claims require an ADR and elevated review. An adapter may not call itself conformant until it passes the published suite.

14.5 Maturity labels

Label	Meaning
EXPERIMENTAL	Interface or semantics may change; no production reliance claim
REFERENCE	Implements frozen public contract and passes reference tests
PILOT	Used in controlled institutional setting with documented limits
CONFORMANT	Passes independent published conformance suite for named version/profile
DEPRECATED	Supported for migration only; replacement identified

15. Alternatives Considered

15.1 Alternative A - Direct document-to-Rego

Advantages	Costs / risks
Fastest demo; few components; easy runtime story	Target lock-in; weak authority; no admission boundary; poor ambiguity preservation; unsupported fields; difficult amendment impact

Decision: Rejected as canonical architecture. Retained as an implemented benchmark baseline and optional unsafe/experimental mode that cannot publish production envelopes.

15.2 Alternative B - RAG plus human-authored controls

Advantages	Costs / risks
Low automation risk; familiar workflow; humans retain responsibility	Manual bottleneck remains; repeated target-specific interpretation; weak reusable meaning; limited change analysis

Decision: Retained as a fallback workflow and baseline. OIC may export source-grounded review packs for human-only authoring, but this does not satisfy the product promise.

15.3 Alternative C - Knowledge graph as source of truth

Advantages	Costs / risks
Strong relations, queries, visualization, extensibility	Graph presence does not establish admission; executable semantics uncertain; state dimensions can collapse; target compilation unresolved

Decision: Use graph structures internally, but canonical meaning is the typed admitted IR with explicit state and schemas, not an unconstrained graph database.

15.4 Alternative D - Full LegalRuleML/Akoma Ntoso implementation

Advantages	Costs / risks
Standards alignment; rich legal representation; interoperability	Large XML/ontology burden; enterprise documents poorly fit; week-one scope impossible; practitioner UX and runtime transformation still unsolved

Decision: Borrow concepts and provide future mappings/profiles. Do not make complete standards compliance a v0.1 dependency.

15.5 Alternative E - Microservices from day one

Advantages	Costs / risks
Independent scaling/security/ownership	Distributed complexity, slow integration, operational burden, premature boundaries

Decision: Modular monolith with isolated untrusted workers and explicit contracts. Extract services only under measured triggers.

15.6 Alternative F - Human approval as a generic checkbox

Advantages	Costs / risks
Simple UX; familiar workflow	Does not establish reviewer standing, scope, source/version, qualification, or correction; becomes liability sink

Decision: Rejected. Admission is a structured institutional event, not a generic approval log.

16. Seven-Day Experimental Reference Slice

The seven-day objective is an experimental, open, reproducible vertical slice. It is not a full enterprise product, full OIC-Bench run, validated usability study, accessibility certification, independent security review, air-gapped production deployment, or evidence of commercial superiority.

The slice uses one prepared bounded corpus and the preflight subset in Section 10.1.1. Any item not completed remains visible as an open gate; scope is reduced rather than rhetorically declared complete.

Day	Deliverables	Exit criterion
Day 0 - Owner freeze	Approve v1.1, requirement IDs, draft schemas, preflight corpus provenance, claims register, prohibited claims, dependency status, and reference hardware	Exploratory code may start; release-freeze and superiority claims remain blocked
Day 1 - Contracts and ingestion	Repository, CI, SourceDocument/SourceAnchor/Candidate/Admission/IR/Envelope schemas, Docling sandbox, hashing, prepared corpus manifest	Prepared source files parse to stable anchors; draft schemas validate fixtures
Day 2 - Authority and extraction	Minimal Authority Registry, source-status workflow, structured extractor, field-level support validator, model/config manifest	Candidate units contain exact source support; no candidate is executable
Day 3 - IR, provisional warrant, and docket	Canonical IR, state validation, provisional ZTL adapter or transparent reference evaluator, dependency graph, ambiguity/conflict detection, prioritization	Unknown and contradiction remain visible; blocking questions are produced
Day 4 - Review and admission	Review console, role checks, disposition workflow, immutable AdmissionRecord, source and impact view	Internal nontechnical test user completes the prepared workflow; this is not a formal usability study
Day 5 - Envelope and OPA	Open Control Envelope, human control matrix, Rego adapter, target tests, POST /admit	A reviewed source clause controls a live request with exact lineage
Day 6 - Provisional continuity and change	Transparent lifecycle/evidence contract, provisional VEIP adapter, replay fixture, amendment impact, kill switches, Docker Compose, security fixtures	A source amendment changes affected artifact state and runtime behavior in the prepared scenario
Day 7 - Preflight and experimental tag	Run preflight metrics and named technical baselines on the reduced corpus; fresh fixture smoke test; SBOM; signed experimental tag; limitations; open gates; roadmap	Reproducible experimental artifact ships only if critical invariants pass; otherwise no tag or narrower scope

16.1 Definition of done for the experimental slice

- Upload the prepared heterogeneous corpus and recover clause, table, and cell anchors.
- Record bounded source authority status and expose at least one expired delegation and one conflict.
- Extract candidate authority, obligations, conditions, exceptions, evidence, and judgment boundaries.
- Generate a prioritized review docket and record scoped admission.
- Produce canonical IR and either a pinned provisional warrant or explicit CANNOT/reference-evaluator artifact.
- Generate a portable Open Control Envelope and human control matrix.
- Compile the deterministic subset to Rego and run generated tests.
- Evaluate a live request and return ALLOW, DENY, CANNOT, ESCALATE, or NOT_APPLICABLE.
- Trace the result to exact source, authority, interpretation, reviewer, warrant/evaluator, envelope, and target versions.
- Introduce an amendment or revocation, invalidate affected controls, re-review, and change the runtime result.
- Run the public preflight subset and publish raw results, failures, limitations, environment, and commands.
- Run locally from a signed experimental release and inspect the same semantic artifacts in the hosted Lab shell if the Lab is included.

16.2 Explicit week-one exclusions

- Full OIC-Bench v0.1, private held-out evaluation, statistical generalization, or superiority claims.
- Formal Fortune 500 practitioner usability study or WCAG 2.2 AA certification.
- Independent security audit, penetration test, or enterprise compliance certification.
- Production on-prem or air-gapped customer deployment.
- Universal legal ontology or full cross-jurisdictional authority resolution.
- Every document format, language, runtime, and domain.
- Autonomous legal interpretation or production compliance certification.
- Full Akoma Ntoso/LegalRuleML authoring environment.
- Catala/OpenFisca compilers or a new policy execution engine.

- Production ingestion of confidential or regulated data in the public Lab.
- Webhook-driven customer production deployment.
- High-availability multi-region enterprise operations.

16.3 Thirty-, sixty-, and ninety-day gates

- **Thirty days:** reproducible alpha, verified dependency dossiers or documented replacements, security preflight, hosted Lab alpha, broader public corpus, and external technical review.
- **Sixty days:** full benchmark, private held-out run, named baseline reports, real practitioner usability study, accessibility audit, and external security review sufficient for scoped public quality claims.
- **Ninety days:** controlled enterprise pilot candidate with SSO, tenancy, on-prem profile, operational runbooks, SLO evidence, customer-specific authority workflow, and signed deployment/evidence boundaries.

17. Limitations and Capability Roadmap

17.1 v0.1 limitations

- OIC does not determine whether an admitted interpretation is legally correct.
- OIC does not replace authorized legal, policy, compliance, audit, or regulatory review.
- Authority Registry quality depends on supplied source metadata and steward decisions.
- Open-textured provisions remain judgment procedures unless the institution admits an evaluation method.
- Complex scans, handwriting, diagrams, and merged tables may require source correction.
- Cross-jurisdictional precedence is not fully automated.
- Multilingual semantic equivalence is outside v0.1 release gates.
- Rego is the primary executable target; DMN is limited to suitable decisions; Cedar is later.
- Runtime enforcement can only evaluate observable and trustworthy facts supplied by integrated systems.
- A control envelope is only as current as its source registry, authority state, evidence, and change feed.
- Hosted Lab launch is not approved for confidential or regulated production data.
- Benchmark results are bounded to named corpora and do not establish universal superiority.

17.2 Capability maturity roadmap

Stage	Capabilities	Evidence required to advance
Stage 1 - Source-to-control reference	Ingestion, anchors, candidate extraction, authority registry, docket, admission, envelope, Rego, tests, lineage	Open release; frozen benchmark; reproducible demo; limitations
Stage 2 - Enterprise corpus integrity	Amendments, supersession, stronger authority graphs, multilingual, tables/annexes, on-prem, SSO, tenant controls	Design partner results; security review; corpus scale; usability
Stage 3 - Continuous institutional compilation	Source monitoring, impact invalidation, re-admission, controlled publication, CI/CD, historical replay	Change-impact benchmark and operational incident tests
Stage 4 - Dependable regulated agents	Agent identity/mandate, dynamic grounding, ZTL warrants, VEIP lifecycle, delegation/revocation, reliance/correction, multi-agent chains	Real runtime pilot; measured consequence control; independent audit
Stage 5 - Open institutional infrastructure	Interoperable envelope standard, independent compilers, domain packs, conformance certification, federated authority graphs	Independent implementations and community governance

17.3 Advancement gates

- No stage is declared complete solely from feature existence; named benchmark and operational evidence are required.
- Enterprise production claims require security, reliability, data-governance, and on-prem evidence in addition to semantic metrics.
- Dependable-agent claims require runtime, reliance, revocation, and correction demonstrations - not only document compilation.
- Open-standard claims require at least one independent implementation or consuming project.

18. Risks, Dependencies, External Review, and Approval

18.1 Principal risks

Risk	Impact	Likelihood before evidence	Mitigation	Owner
R-01 Semantic hallucination becomes executable	Critical	High	Source support, candidate/admission separation, unsupported-field gate	Semantic lead
R-02 Wrong source treated as authority	Critical	High	Authority registry, role-bound admission, conflict docket	Authority steward
R-03 Seven-day narrative outruns actual delivery	High	High	Experimental-slice scope, visible open gates, no full-benchmark claim	Design authority
R-04 ZTL or VEIP cannot be independently verified	High	Medium-high	Mandatory dossiers, provisional adapters, transparent fallback contracts, ADR reopen	Integration lead
R-05 Reviewer approves without standing or sufficient context	Critical	Medium	Reviewer authority record, scoped disposition, separation of powers	Product/security lead
R-06 Benchmark overfits favorable corpus	High	Medium	Held-out/adversarial partitions, contamination log, external adjudication	Benchmark lead
R-07 Target adapter changes meaning	Critical	Medium	Canonical tests, source maps, cross-target equivalence, release block	Compiler lead
R-08 Community fragments schema through incompatible PRs	Medium	Medium	Schema governance, profiles, conformance suite, ADRs	Maintainers
R-09 Public claims outrun evidence	High	High	Claims ledger, evidence classification, forbidden register, release approval	Design authority
R-10 Commercial differentiation is weaker than assumed	High	Medium-high	Product trials, capability matrix, customer discovery, narrow claims	Product lead
R-11 Enterprise buyer expects autonomous legal compliance	High	High	Product language, limitations, review workflow, contract scope	Product lead
R-12 Runtime depends on unavailable evidence systems	High	High	Observable-input analysis, CANNOT/ESCALATE, integration roadmap	Enterprise architect
R-13 Public Lab receives confidential or malicious documents	Critical	High	Public-only restriction, upload controls, sandboxing, deletion, warning	Security lead
R-14 Numerical targets create false precision	Medium	High	Evidence-class labels, calibration ADRs, raw results	Benchmark/finance owners

18.2 External dependencies

Dependency	Outside control	Current status	If delayed or changed
Docling and parsing ecosystem	Upstream versions/vulnerabilities	Public dependency; pin pending implementation freeze	Adapter isolation; fallback parser for bounded formats; block unsafe source
Model providers/local models	Availability, terms, quality, cost	Provider-neutral design; models not frozen	Provider abstraction; local profile; benchmark before update; pause extraction
ZTL	Project-controlled artifacts, interface, license, reproduction	Provisional; dossier incomplete	Use transparent reference evaluator or review-only mode; no fabricated warrant or maturity claim
VEIP	Project-controlled specification, implementation, license, evidence lifecycle	Provisional; dossier incomplete	Emit open lifecycle/evidence contract; defer full adapter; do not hide dependency gap
OPA/Rego	Version and behavior	Public dependency; pin pending	Pin engine; conformance suite; adapter replacement
Benchmark source rights	Licenses and redistribution	Corpus not yet frozen	Use public/synthetic corpus or publish hashes/scripts where redistribution is prohibited
Enterprise identity/evidence systems	Customer integrations	Mocks only in reference slice	Explicit non-production state until connected
Domain, legal, security, accessibility	Availability and independence	Not yet complete	Corresponding claims and release states

Dependency	Outside control	Current status	If delayed or changed
reviewers			remain blocked
Commercial product access	Trial availability and confidential product detail	Public-materials review only	Mark capability unknown rather than absent; no unsupported competitor claim

18.3 Anticipated objections

The following are internally generated design challenges, not evidence of independent dissent:

1. **"Human admission recreates the bottleneck."** OIC does not eliminate judgment; it attempts to concentrate review on blocking questions and measure active time. The objection remains valid if usability and quality data fail.
2. **"Authority cannot be normalized across enterprises."** The registry is a scoped enterprise profile, not a universal legal hierarchy. The objection remains valid if extension requirements fracture the core model.
3. **"A knowledge graph plus RAG is sufficient."** The benchmark must compare a graph baseline on admission, unsupported fields, change impact, and executable equivalence.
4. **"The envelope is another proprietary schema."** Schema, conformance tests, examples, and adapters are intended to be open. The objection remains valid until independent consumption occurs.
5. **"The commercial market already provides this."** Public materials show significant overlap. OIC must prove a narrower open, source-anchored, admission-gated, portable transformation rather than dismissing existing products.
6. **"Benchmark superiority can be manufactured by choosing favorable dimensions."** Shared metrics and baseline capabilities must be reported; OIC-specific dimensions remain separate; external reproduction is required for mature claims.

18.4 External review and dissent register

Review ID	Reviewer status	Finding	Disposition	Residual risk / close condition
ER-OIC-001	External critical review supplied 28 July 2026; reviewer identity not recorded in this artifact	Seven-day plan is not credible against full benchmark/security/usability scope	Accepted; scope split across 7/30/60/90-day horizons	Close after measured delivery history supports a revised schedule
ER-OIC-001	Same	ZTL and VEIP are load-bearing but unverifiable	Accepted; provisional status and mandatory dossiers added	Close only with pinned public/project artifacts, licenses, conformance, and independent status
ER-OIC-001	Same	Standards-body register exceeds actual approval maturity	Accepted; status downgraded to owner-approved pre-external-review	Close when named external approvals in 18.5 are complete
ER-OIC-001	Same	Commercial landscape omitted	Accepted; Section 3.2 added from public materials	Close after hands-on or independently sourced product evaluation
ER-OIC-001	Same	Thresholds and cost figures appear sourced but are not	Accepted; evidence classification added	Close after calibration and measured reports
ER-OIC-001	Same	Dissent register is simulated	Accepted; renamed anticipated objections; this table starts actual review record	Close after additional named independent reviewers participate
ER-OIC-001	Same	Aspirational tone conflicts with technical register	Accepted in part; founder thesis retained but made explicitly non-normative	Public launch copy still requires claims review

External objections **MUST** be preserved substantially as raised, including unresolved disagreement. The design authority may decide despite dissent, but must record the reason, evidence, residual risk, and revisit trigger.

18.5 Decision authority and required approvals

Decision area	Decision-maker / approver	Current status
Product problem, architecture hypothesis, exploratory scope	Arkadiy Miteiko / Veraxis	Owner-approved for experimental implementation
Technical architecture and interfaces	Named technical lead plus independent senior engineer	Owner-approved; independent review pending
ZTL semantic boundary and dependency dossier	ZTL owner/reviewer plus OIC integration lead	Required before release freeze; pending
VEIP semantic boundary and dependency dossier	VEIP design authority plus OIC integration lead	Required before release freeze; pending
Benchmark corpus, annotations, metrics, and thresholds	Benchmark lead plus independent adjudicator and design authority	Proposed; corpus and calibration pending
Security and public Lab scope	Security lead plus independent reviewer and design authority	Public-only limitation selected; technical review pending
Accessibility	Qualified accessibility reviewer	Pending before evidence release

Decision area	Decision-maker / approver	Current status
Licensing	Design authority plus counsel	Proposed baseline; counsel review pending
Commercial comparison language	Product lead plus claims reviewer	Public-materials matrix approved as provisional; product diligence pending
Public quality claims	Design authority plus benchmark approver	Blocked until frozen evidence passes

18.6 Conditions for code start and release states

Exploratory code start

- TDD-OIC-001 v1.1 is checked in as owner-approved pre-external-review.
- Requirement IDs and invariants are checked into the repository.
- Draft schemas and dependency status are explicit.
- Preflight corpus rights and provenance are documented.
- CLAIMS.md distinguishes evidence, hypotheses, targets, and forbidden claims.
- Public Lab restrictions are visible before upload.
- Each implemented module has a named owner.

Experimental tag

- Critical invariants pass on the prepared corpus.
- Preflight metrics, failures, raw outputs, commands, environment, and limitations are published.
- ZTL/VEIP adapters are labeled provisional if dossiers remain incomplete.
- SBOM and signed release artifacts exist.
- No enterprise-readiness or superiority language appears.

Release freeze

- Required independent reviews in 18.5 are complete.
- ZTL/VEIP dossiers pass or replacement ADRs are accepted.
- Full benchmark corpus, annotations, baselines, and thresholds are frozen.
- Security, privacy, accessibility, and licensing gates pass for the declared profile.
- Claims ledger contains only evidence-supported statements.

19. Verification, Release, and Baseline Use

This TDD controls owner-authorized exploratory implementation and the evidence required to advance release status. It does not itself prove that the design is correct, novel, enterprise-ready, secure, or commercially differentiated. Development work **MUST** cite relevant requirement IDs and ADRs. A feature that cannot be traced to the TDD is out of scope or requires an approved change.

19.1 Pull-request requirements

Every material PR states:

- customer problem and requirement IDs served;
- modules and interface versions changed;
- build/reuse decision and dependency effect;
- tests, fixtures, benchmark partition, and evidence-class changes;
- security/privacy impact;
- failure and rollback behavior;
- claims and limitations impact;
- whether hosted/open semantic parity changes;
- whether ZTL/VEIP provisional boundaries or dossiers change.

19.2 Release evidence pack

Each tagged release contains:

- source commit, signed tag, build provenance, SBOM, and dependency lockfile;
- exact schemas and migration status;
- compiler, model, prompt, parser, target, ZTL/VEIP or fallback versions;
- corpus and annotation manifests appropriate to the release stage;
- raw benchmark or preflight results and baseline reports;
- security, privacy, and accessibility status;
- known limitations, unsupported scenarios, and failed gates;
- claims ledger showing evidence class for every public statement;
- release decision, reviewers, unresolved dissent, residual risks, and rollback plan.

19.3 Falsification conditions

The design's central assumptions are materially weakened if evidence shows any of the following:

- practitioners cannot reach useful admitted controls without review effort comparable to existing manual work;
- source anchoring or authority reconstruction is unreliable enough that unsupported controls remain common;
- direct document-to-code or commercial alternatives match OIC on source support, ambiguity, change impact, portability, and review with less complexity;
- the canonical IR cannot represent real enterprise variation without uncontrolled fragmentation;
- target compilation repeatedly changes admitted meaning;
- ZTL or VEIP cannot be verified, replaced, or integrated without hidden semantic dependence;
- policy amendments cannot reliably identify affected controls and reliance;
- hosted/open parity cannot be reproduced;
- customers do not value a portable institution-owned control envelope;
- the security and privacy cost of processing institutional documents exceeds the practical benefit.

These outcomes require redesign, narrowing, or termination. They may not be reframed as success through selective metrics.

19.4 Baseline rule

TDD-OIC-001 v1.1 is the owner-approved, pre-external-review baseline for the experimental implementation. It becomes approved for implementation or frozen for release only through the explicit gates in Sections 10 and 18. No document label, normative verb, architecture diagram, or founder approval substitutes for external evidence.

Appendix A. Requirements Traceability Matrix

ID	Requirement	Responsible modules	Verification	Primary metric
CUS-001	Convert existing institutional documents into runtime-enforceable controls	M1-M10	Reference E2E flow	Time to first envelope
CUS-002	Reduce manual interpretation/reimplementation across systems	M5, M9, M10	Multi-target projection	Cross-target equivalence
CUS-003	Allow nontechnical practitioners to review results	M7, M8	Usability study	Orientation and review time
CUS-004	Support deployment within days	All reference modules	Seven-day release demo	Elapsed delivery and setup time
CUS-005	Keep institution-owned artifacts portable	M5, M9, M10	Local export/import	Independent target adapter
SRC-001	Preserve original file bytes and hashes	M1	Immutable-source fixture	Hash match
SRC-002	Anchor every candidate field to source	M2, M4	Anchor audit	Anchor precision/recall
SRC-003	Preserve tables, cells, definitions, annexes, references	M1, M2	Structured corpus tests	Structure coverage
SRC-004	Represent amendments and supersession	M2, M3	Version-pair tests	Change-impact recall
AUTH-001	Record authority basis and scope	M3	Gold authority corpus	Authority F1
AUTH-002	Represent delegation/subdelegation and limits	M3	Delegation fixtures	Delegation accuracy
AUTH-003	Prevent expired/revoked authority from ALLOW	M3, M9, M12	Expiry/revocation tests	Recall and runtime correctness
SEM-001	Extract typed normative units	M4	Gold normative annotations	Macro-F1
SEM-002	Preserve alternatives and open texture	M4, M7	Ambiguity corpus	Ambiguity recall
SEM-003	No silent default values	M4, M5	Missing-field mutations	False-resolution rate
IR-001	Canonical target-independent IR	M5	Round-trip/multi-target tests	Equivalence
IR-002	Separate interpretation, epistemic, lifecycle states	M5	State-machine tests	Invariant pass
IR-003	Deterministic canonical hashing	M5	Canonicalization fixtures	Hash reproducibility
ZTL-001	Preserve T/F/Z and CANNOT	M6	ZTL fixtures	Unknown-to-false = 0
ZTL-002	Bind warrant to exact grounds/formula/version	M6	Replay tests	Warrant hash match
REV-001	Generate prioritized blocking docket	M7	Gold issue set	Recall and compression
REV-002	Record scoped institutional admission	M8	Role/disposition tests	Record completeness
REV-003	Separate reviewer, publisher, correction roles	M8, M12	Authorization tests	No privilege bypass
ENV-001	Generate portable Open Control Envelope	M9	Schema/conformance tests	Field coverage
ENV-002	Require explicit failure behavior	M9	Missing/unknown/error cases	No silent ALLOW
ENV-003	Support deterministic and human-judgment controls	M9	Mixed corpus	Classification accuracy
TGT-001	Compile to OPA/Rego	M10	OPA compile/tests	Compile and conformance
TGT-002	Provide DMN export for suitable subset	M10	DMN validation	Export correctness
TGT-003	Do not mutate meaning for target fit	M10, M11	Mutation/equivalence	Mismatch rate
TST-001	Generate comprehensive behavioral tests	M11	Gold cases	Behavioral conformance
TST-002	Run semantic mutation tests	M11	Mutation suite	Survivor report
CHG-001	Identify affected controls after source change	M2-M7, M11	Version-pair corpus	Change-impact recall
RUN-001	Return structured operational	M12	Runtime API tests	Correct response schema

ID	Requirement	Responsible modules	Verification	Primary metric
	verdict			
RUN-002	Preserve evidence, replay, reliance, correction	M12	VEIP lifecycle tests	Replay/correction integrity
SEC-001	Sandbox untrusted documents and model output	M1, M4	Adversarial security corpus	No escape/tool action
SEC-002	Prevent cross-tenant leakage	Platform	Authorization/security tests	Zero leakage findings
SEC-003	Support local/on-prem/offline mode	Deployment	Network-deny test	Zero outbound dependency
OPS-001	Provide kill switches and rollback	Platform/M12	Game-day tests	Recovery success
BEN-001	Freeze corpus, thresholds, and baselines pre-claim	M11/OIC-Bench	Manifest checks	Reproducibility
BEN-002	Disclose limitations and forbidden claims	Repository/Lab	Release audit	Claims compliance
OSS-001	One-command local reference run	Repository	Clean-machine test	Setup success/time
OSS-002	Hosted and open compiler parity	Platform/core	Artifact diff test	Semantic parity
A11Y-001	Keyboard-accessible practitioner workflow	UI	Accessibility test	WCAG/task completion

Appendix B. Metric Definitions and Evaluation Rules

B.1 Source-supported executable fields

For each executable envelope field, annotators determine whether the field is supported by exact source anchors plus any explicit admitted derivation. The numerator is supported executable fields; the denominator is all executable fields. Merely citing a nearby paragraph does not count if the source does not support the actor, modality, threshold, exception, evidence, or time represented.

B.2 Unsupported executable-field rate

An executable field is unsupported when no defensible source/admission path exists, the cited source contradicts it, or an inference was not explicitly admitted. Critical authority/permission inventions are separately reported and may fail release regardless of aggregate percentage.

B.3 Normative Field macro-F1

Calculate precision, recall, and F1 separately for actor, action, object/resource, modality, conditions, exceptions, evidence duties, temporal scope, escalation/review, and discretion. Report macro-average and every field. Matching rules and partial-credit policy are frozen with annotations.

B.4 Authority Reconstruction F1

Gold edges include issued_by, derives_authority_from, superior_to, delegates_to, permits_subdelegation, implements, interprets, amends, supersedes, conflicts_with, effective_during, and scoped_to. Evaluate entity/edge match and material attributes.

B.5 Ambiguity recall and false resolution

A gold blocking ambiguity is detected only when the system exposes an actionable question or explicit unresolved state before envelope publication. A false resolution occurs when the system emits a determinate executable meaning despite gold ambiguity/unknown without an admission resolving it.

B.6 Behavioral conformance

Run gold transactions against the admitted canonical evaluator and target engine. Compare operational verdict and required reason class. Cases include positive, negative, boundary, exception, missing evidence, unknown, expired, revoked, conflict, and unauthorized actor.

B.7 Change-impact recall

For each source-version pair, gold annotations identify materially affected definitions, interpretations, authority edges, controls, tests, and runtime artifacts. The system passes only when it identifies the full required review/invalidation set within the defined level of granularity.

B.8 Usability

Measure orientation time, active task time, task completion, material error rate, reviewer agreement, confidence calibration, requests for technical help, and System Usability Scale or equivalent. Report participant roles and prior experience. Small samples are described as formative, not statistically generalizable.

B.9 Operational measurements

Latency is measured on declared hardware with cold/warm state defined. Model cost includes all tokens, retries, repair calls, and provider charges. Human time excludes queue/scheduling delay but includes active source correction and review. All scripts and raw aggregate data are released where privacy permits.

Appendix C. Claims and Forbidden-Claims Register

ID	Claim	Required evidence	Status
CLM-001	OIC converts heterogeneous policy documents into source-anchored candidate normative units.	Release demo + anchor metrics	Permitted after gates pass
CLM-002	OIC separates extracted meaning from institutionally admitted meaning.	State/admission tests	Permitted
CLM-003	OIC preserves unresolved ambiguity rather than silently forcing certainty.	Ambiguity/false-resolution metrics	Permitted after gates pass
CLM-004	OIC produces a portable target-independent Open Control Envelope.	Schema + two projections over roadmap	Permitted for reference envelope; target independence qualified until second target
CLM-005	Every published runtime verdict is traceable to source, authority, admission, control, and version.	Lineage/replay tests	Permitted after full chain passes
CLM-006	OIC identifies affected controls after source amendments.	Change-impact benchmark	Permitted within named corpus
CLM-007	OIC outperforms direct document-to-code approaches on named dimensions.	Frozen baseline deltas	Permitted only with exact baseline/corpus/metrics
CLM-008	OIC is the first institutional compiler.	Independent prior-art review and defensible definition	Not permitted at v0.1 without external verification
CLM-009	OIC guarantees legal compliance.	Impossible from current scope	Forbidden
CLM-010	OIC eliminates legal/policy review.	Contradicts design	Forbidden
CLM-011	OIC automatically understands institutional intent.	Not measurable/established	Forbidden
CLM-012	OIC supports all enterprise policies and jurisdictions.	Not true	Forbidden
CLM-013	OIC turns agents into dependable actors.	Requires Stage 4 runtime/reliance/correction evidence	Roadmap thesis only
CLM-014	The open repository and hosted Lab use the same semantic compiler core.	Commit/config parity test	Required claim/invariant

Appendix D. Release Checklists

D.1 Technical release

- ☐ Signed immutable release tag exists and matches published commit.
- ☐ Schemas and examples validate; migrations documented.
- ☐ Dependency versions pinned; SBOM/provenance/signatures generated.
- ☐ All invariant, unit, integration, security, mutation, and benchmark gates pass or release scope is reduced.
- ☐ Clean-machine Docker Compose demo succeeds.
- ☐ Hosted Lab compiler commit/config is disclosed and parity-tested.
- ☐ OPA artifacts compile and runtime examples reproduce.
- ☐ Amendment, revocation, replay, and correction demo passes.
- ☐ No secrets, customer content, or unlicensed corpus material included.

D.2 Documentation and claims

- ☐ README states customer problem before architecture.
- ☐ LIMITATIONS.md is visible from README and Lab results.
- ☐ CLAIMS.md maps every statement to evidence and scope.
- ☐ ROADMAP.md separates current, planned, and thesis capabilities.
- ☐ Architecture diagrams and module contracts match code.
- ☐ Benchmark report includes failures, corpus limits, contamination, and costs.
- ☐ Public language avoids legal-compliance, autonomous interpretation, and universal-superiority claims.

D.3 Public Lab

- ☐ Upload screen prohibits confidential, regulated, privileged, export-controlled, and classified content.
- ☐ Retention/deletion and model-provider terms are displayed before upload.
- ☐ Keyboard-only test passes all lifecycle states, drawers, dialogs, and review actions.
- ☐ Candidate/admitted/qualified/rejected/non-automatable states are visually and textually distinct.
- ☐ Every result displays compiler/model/config versions and limitations.
- ☐ Delete operation removes project artifacts according to declared policy.
- ☐ Rate, page, file, token, and cost limits enforced.

Appendix E. Reference Sources, Commercial Materials, and Standards

The following sources informed the design. They are not endorsements of OIC and do not imply standards conformance, product absence, or comparative superiority. Open standards and official product materials were reviewed in July 2026. Commercial capability statements in Section 3.2 reflect only public materials and must be verified through product diligence before procurement or competitive claims.

E.1 Open standards, research, and open-source foundations

1. Docling project and repository: <https://github.com/docling-project/docling>
2. Open Policy Agent documentation: <https://www.openpolicyagent.org/docs>
3. OPA policy language and testing: <https://www.openpolicyagent.org/docs/policy-language> and <https://www.openpolicyagent.org/docs/policy-testing>
4. OASIS Akoma Ntoso / LegalDocML: <https://www.oasis-open.org/standard/akn-v1-0/>
5. OASIS LegalRuleML: <https://www.oasis-open.org/committees/legalruleml/>
6. European Legislation Identifier: <https://eur-lex.europa.eu/eli-register/about.html>
7. New Zealand Better Rules for Government Discovery Report: <https://www.digital.govt.nz/dmsdocument/95-better-rules-for-government-discovery-report/html>
8. OECD, Cracking the Code: Rulemaking for Humans and Machines: <https://www.oecd.org/gov/regulatory-policy/cracking-the-code-rulemaking-for-humans-and-machines.htm>
9. Catala: <https://github.com/CatalaLang/catala>
10. OpenFisca: <https://openfisca.org/>
11. Object Management Group, Decision Model and Notation 1.5: <https://www.omg.org/spec/DMN/1.5>
12. Cedar policy language: <https://github.com/cedar-policy/cedar>
13. W3C PROV-O: <https://www.w3.org/TR/prov-o/>
14. W3C Web Annotation Data Model: <https://www.w3.org/TR/annotation-model/>
15. W3C Shapes Constraint Language: <https://www.w3.org/TR/shacl/>
16. W3C RDF Dataset Canonicalization: <https://www.w3.org/TR/rdf-canon/>
17. NIST Secure Software Development Framework: <https://csrc.nist.gov/Projects/ssdf>
18. NIST AI Risk Management Framework - Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
19. OWASP Prompt Injection: <https://owasp.org/www-community/attacks/PromptInjection>
20. SLSA specification: <https://slsa.dev/spec/>
21. Sigstore Cosign: <https://docs.sigstore.dev/cosign/>

E.2 Commercial and adjacent official materials

22. ServiceNow Policy and Compliance Management: <https://www.servicenow.com/products/policy-compliance-management.html>
23. ServiceNow product documentation for Policy and Compliance Management: https://www.servicenow.com/docs/r/governance-risk-compliance/policy-and-compliance-management/r_PolicyComplianceMgmt.html
24. IBM OpenPages Regulatory Compliance Management: <https://www.ibm.com/products/openpages/regulatory-compliance>
25. IBM OpenPages Policy Management: <https://www.ibm.com/products/openpages/policy>
26. CUBE automated regulatory intelligence: <https://cube.global/>
27. CUBE policy and controls mapping: <https://cube.global/sectors/requirement/policy-and-controls-mapping>
28. LogicGate Policy and Procedure Management: <https://www.logicgate.com/platform/applications/policy-procedure-management/>
29. LogicGate Regulatory Compliance Management: <https://www.logicgate.com/solutions/regulatory-compliance-management/>
30. Norm Ai product and resources: <https://www.norm.ai/>
31. Norm Ai, Legal Engineering Automation Platform description: <https://www.norm.ai/resources/legal-engineering-a-paradigm-shift-in-law>
32. OneTrust AI Governance: <https://www.onetrust.com/solutions/ai-governance/>
33. OneTrust AI Runtime Controls: <https://www.onetrust.com/solutions/ai-governance/runtime-controls/>
34. Credo AI product: <https://www.credo.ai/product>
35. Credo AI Agent Governor: <https://www.credo.ai/agent-governor>

E.3 Project-controlled dependency records

36. ZTL project artifacts and pinned interfaces: **not yet accepted as a release dependency**. Required dossier fields are specified in Section 3.5. The repository location, license, interface version, fixture digest, conformance evidence, and independent reproduction status must be added before release freeze.
37. VEIP specifications and Authorization Evidence Pack: **not yet accepted as a release dependency**. Required dossier fields are specified in Section 3.5. The repository/specification location, license, interface version, lifecycle fixtures, conformance evidence, and independent reproduction status must be added before release freeze.
38. External Review ER-OIC-001, 28 July 2026: supplied critique incorporated into v1.1; source retained in the project evidence room. Reviewer identity and independent affiliation are not recorded in this document and therefore must not be implied.

Appendix F. Decision Log Seed

ADR	Decision	Status
ADR-001	Use a modular monolith for reference implementation	Accepted
ADR-002	Canonical meaning resides in Institutional IR, not target code	Accepted
ADR-003	Open Control Envelope is the portable enforcement boundary	Accepted as design hypothesis
ADR-004	OPA/Rego is the first executable target	Accepted
ADR-005	Hosted Lab uses the same open semantic core	Accepted
ADR-006	Public Lab accepts only public, synthetic, or non-confidential samples	Accepted
ADR-007	Unknown never converts to false; no silent ALLOW	Accepted
ADR-008	Institutional admission is structured and role-bound	Accepted
ADR-009	ZTL validates logical warrant, not source authority	Provisional pending dependency dossier
ADR-010	VEIP preserves consequence lifecycle, not policy meaning	Provisional pending dependency dossier
ADR-011	PostgreSQL JSONB/edges before dedicated graph database	Accepted for reference and pilot
ADR-012	Apache-2.0 code/schema and CC BY 4.0 docs/annotations	Proposed pending counsel
ADR-013	DMN export is secondary, not canonical	Accepted
ADR-014	Cedar adapter deferred until after Rego reference	Accepted
ADR-015	Benchmark thresholds selected before held-out claims	Accepted; values remain proposed pending calibration
ADR-016	Replace full seven-day release narrative with experimental slice and 30/60/90-day evidence horizons	Accepted in v1.1
ADR-017	Downgrade status to owner-approved, pre-external-review	Accepted in v1.1
ADR-018	Treat ZTL and VEIP as provisional project-controlled dependencies until dossiers pass	Accepted in v1.1
ADR-019	Add commercial capability landscape and prohibit absence claims from public documentation alone	Accepted in v1.1
ADR-020	Classify every schedule, capacity, cost, and threshold number by evidence type	Accepted in v1.1
ADR-021	Separate anticipated objections from actual external dissent	Accepted in v1.1
ADR-022	Keep founder thesis as explicitly non-normative front matter	Accepted in v1.1

Appendix G. Final Baseline Statement

TDD-OIC-001 v1.1 supersedes v1.0 as the owner-approved, pre-external-review baseline for Open Institutional Compiler development beginning 28 July 2026. Engineers, reviewers, benchmark owners, security personnel, product teams, and public communicators must use its requirements, invariants, module boundaries, interfaces, evidence classifications, release gates, limitations, and claims controls for the declared experimental scope.

The implementation is successful only when the two artifacts and benchmark make the same proposition observable from different directions:

The open repository demonstrates whether institutional compilation is inspectable and reproducible. The hosted Lab demonstrates whether the transformation is usable by practitioners. OIC-Bench establishes what the current system can and cannot do within a frozen scope. External use and review determine whether the architecture deserves infrastructural and category claims.

Owner approval permits implementation. It does not substitute for the external evidence required to advance status.

Appendix H. Revision 1.1 Change and External Review Record

H.1 Supersession

- Superseded document: TDD-OIC-001 v1.0, 28 July 2026.
- Superseding document: TDD-OIC-001 v1.1, 28 July 2026.
- Reason: reconcile the design with its own falsifiability standard after External Review ER-OIC-001.
- Architecture disposition: retained with targeted governance, evidence, dependency, schedule, competitive, and claims corrections.

H.2 Material changes

1. Changed status from GOVERNING DEVELOPMENT BASELINE to OWNER-APPROVED DESIGN BASELINE - PRE-EXTERNAL-REVIEW.
2. Moved the founder covenant from the cover into an explicitly non-normative note.
3. Added current-evidence status to the executive decision.
4. Split delivery into seven-day experimental, thirty-day reproducible, sixty-day evidence, ninety-day pilot, and post-pilot maturity horizons.
5. Added commercial and adjacent product landscape from official public materials.
6. Reclassified ZTL and VEIP as provisional project-controlled dependencies with required verification dossiers.
7. Split benchmark preflight from full OIC-Bench and prohibited week-one superiority claims.
8. Relabeled thresholds, capacity, schedule, and cost figures by evidence class.
9. Replaced simulated dissent with anticipated objections and an actual external review register.
10. Updated approval and release gates so owner approval cannot be mistaken for external maturity.
11. Added falsification conditions against commercial alternatives and project-controlled dependency failure.
12. Added ADRs documenting each corrective decision.

H.3 Integrity note

Version 1.0 remains an immutable historical record. Version 1.1 is a consolidated successor rather than a silent edit. The source markdown, generated DOCX, final PDF, render set, and SHA-256 digest must be stored with the release evidence pack.